OPERATING INSTRUCTIONS

MODEL 526



ZAMBONI®

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This manual must be kept and stored with the machine.
The manual and all related safety and service materials are available online at www.zamboni.com

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Introduction

This manual has been prepared to acquaint you with the operation of your ZAMBONI® ice resurfacer and to provide important safety information. Read, understand and comply with the information in this manual. All new operators must read this manual before operating the machine for the first time. Periodic review of the manual is essential for all operators.

The care exercised in the operation and maintenance of the ice resurfacer will extend its useful life, ensure highly satisfactory and dependable service at reduced cost and assure safe operation and maintenance.

We thank you for choosing a ZAMBONI® product and want to assure you of our continuing interest in your satisfaction with your ice resurfacer.

This manual should be considered a permanent part of this machine. It should remain with the machine when sold, to provide the next owner / operator with important safety and maintenance information.

All information, illustrations and specifications contained in this manual are based on the latest product information available at the time of printing. The right is reserved to make changes at any time without notice.

ZAMBONI COMPANY LTD.

ZAMBONI and the configuration of the Zamboni® ice resurfacer are registered trademarks of Frank J. Zamboni & Co., Inc.

Date of Purchase _____

Ice Resurfacer Serial Number _____

THIS BOOK WAS LAST REVISED OCTOBER, 2015

Part Number DM-20077

Safety Symbols

Safety Symbols

The safety symbols listed outline basic safety precautions.

READ AND UNDERSTAND ALL SAFETY PRECAUTIONS AND WARNINGS BEFORE OPERATING, OR PERFORMING LUBRICATION AND MAINTENANCE.

Personal Safety

These symbols indicate when your safety or the safety of others may be at risk.

DANGER!



This safety symbol warns of possible personal injury or death.

WARNING!



This safety symbol warns of possible personal injury.

Equipment Damage

CAUTION!



This symbol warns of possible damage that may occur to the machine.

Ice Resurfacer Guards

The safety guards used on the Zamboni ice resurfacer are illustrated in Figures 1.1 and 1.2.

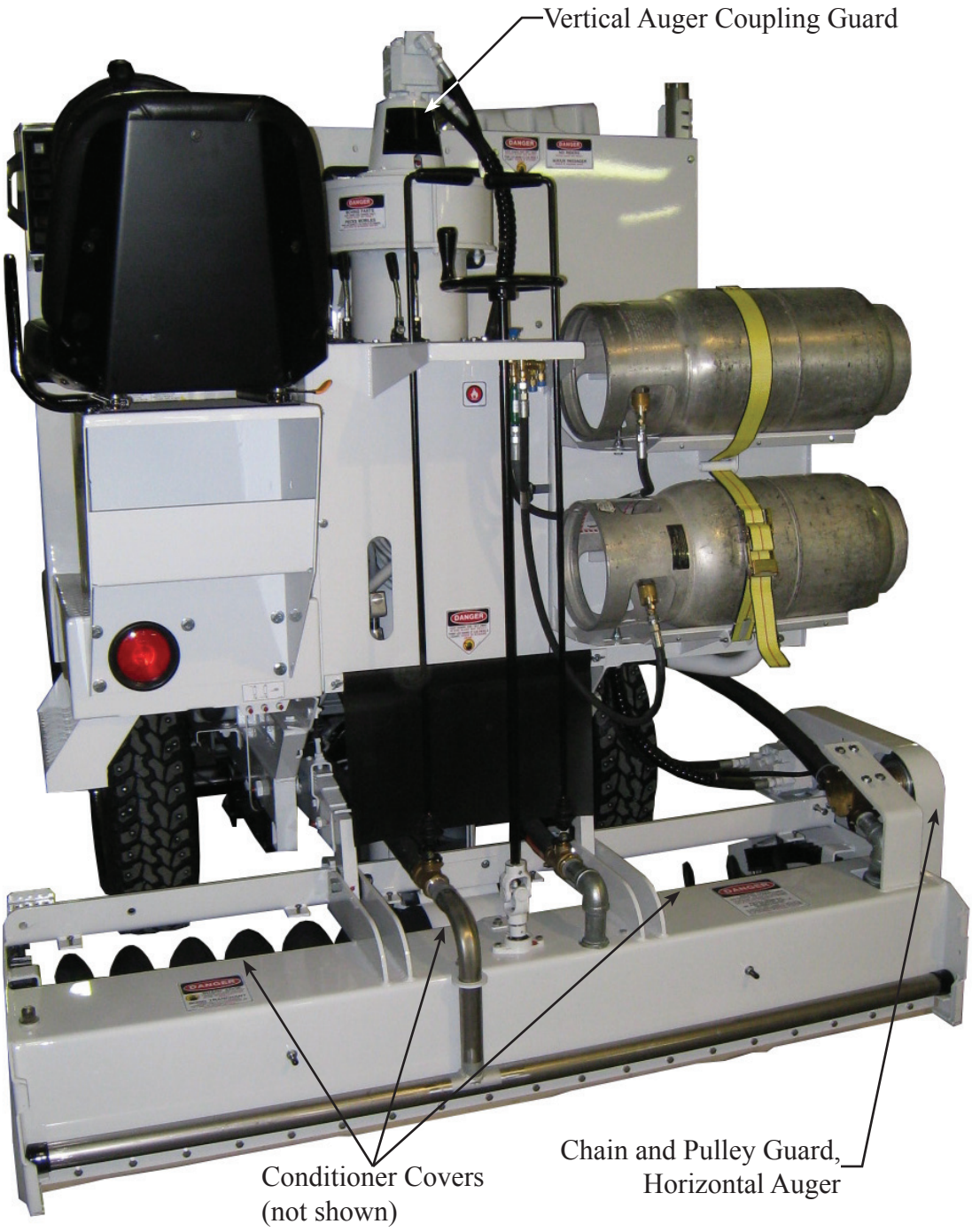


Figure 1.1 *Ice Resurfacer Guards*



Figure 1.2 *Ice Resurfacer Guards - Snow Tank Safety Stand*

Zamboni® Ice Resurfacer Specifications

Engine	Mitsubishi 2.4L SOHC L4 Fuel Types: Gasoline, Propane (LPG), Natural Gas (CNG). Available as dual fuel (Gasoline / Propane, Natural Gas / Propane) or dedicated fuel. Load sensing electronic governor enables true "hands-free" operation by the driver.
Hydrostatic Transmission	Sundstrand variable displacement pump and fixed displacement motor create a full time 4WD hydrostatic system.
Chassis	Tubular steel frame, Dana axles with rear leaf spring suspension.
Tires	Steel belted radial LT215/85R16 load range E rated with tungsten tipped studs.
Hydraulics	Full proportional controls with priority to power steering.
Electrical System	12 VDC system with 70 Amp alternator.
Instruments	CAN driven 3 in one gauge featuring tachometer, oil pressure and coolant temperature. Multi Function CAN unit capable of displaying multiple engine signals as well as service intervals. Overheat, alternator and oil pressure warning lights and audible alarm. Water pump switch, ignition switch, horn button, fuses, gasoline fuel gauge.
Ice Shaving Blade	Driver Adjustable, 195.6 x 12.7 x 1.27 cm (77 x 5 x ½ in.)
Wash Water Pump	12 VDC with integral electro-magnetic clutch.
Conveyor System	Two 25.4 cm (10 in.) diameter double flighted augers (horizontal / vertical).

Tank Capacities

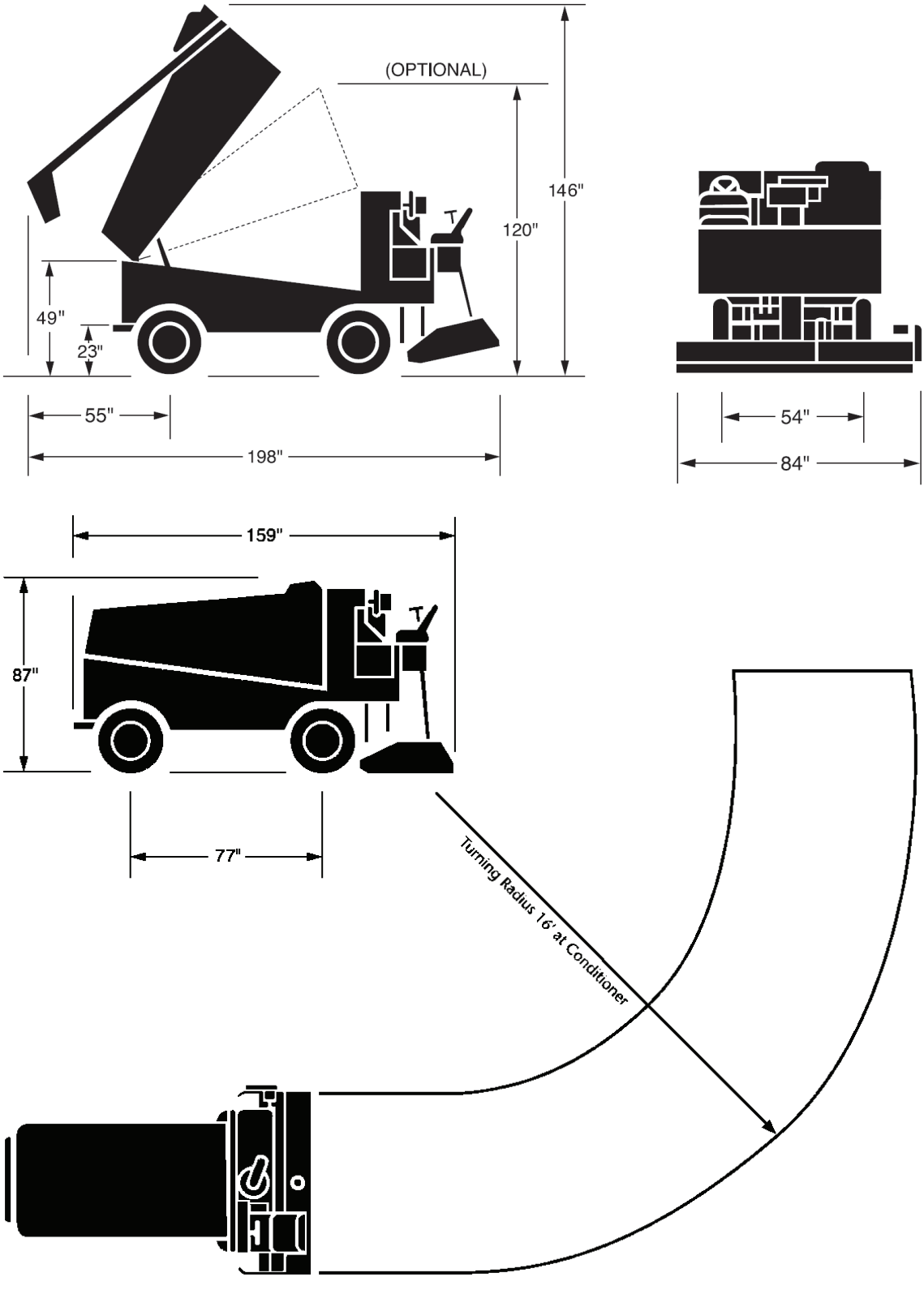
Snow Tank	Actual 2.83 m^3 (100 ft^3)	Compacted 3.54 m^3 (125 ft^3)	
Ice Making Water	727 L	192 USG	160 IMP GAL
Wash Water	273 L	72 USG	60 IMP GAL
Total	1000 L	264 USG	220 IMP GAL
Hydraulic Oil	95 L	25 USG	21 IMP GAL
Gasoline	41.6 L	11 USG	9.2 IMP GAL
Propane (LPG)	30 lb recommended**		
Natural Gas (CNG)	120 L	32 USG	26 IMP GAL

**Tanks are not supplied from the factory.

Dimensions

	Snow Tank Raised	Snow Tank Lowered
Length	5.03 m (198 in.)	4.04 m (159 in.)
Height	3.71 m (146 in.)	2.21 m (87 in.)
Width	2.13 m (84 in.)	
Wheel Base	1.95 m (77 in.)	
Wheel Track	1.37 m (54 in.)	
Turning Radius	4.86 m (16 ft.)	
Weight	Empty: 3143 kg (6930 lb) With Water: 4240 kg (9350 lb)	

Dimensions


(OPTIONAL)
146"
120"
49"
23"
55"
198"
54"
84"
159"
87"
77"
Turning Radius 16' at Conditioner

Foreword

- This manual contains safety, lubrication and maintenance information.
- The manual is a reference for the new operator and a refresher for the experienced one.
- It is mandatory that all operators review in detail this manual before performing any operation.
- **Read - understand - and keep** this manual with the Zamboni ice resurfacer.
- Some illustrations show details or attachments that may be different from your machine. Guards and covers may have been removed for illustrative purposes only.
- **KEEP ALL GUARDS IN PLACE!**
- Continuing improvement and advancement of product design may have caused changes to your Zamboni machine which may not be included in this publication.
- Whenever a question arises regarding your ice resurfacer or this publication, please consult Zamboni Company Ltd. for the latest information.
- A continuing review of this manual is recommended.

Safety

The "General Safety Practices" section lists minimum safety precautions. In addition, this section explains the **warning labels** used on the Zamboni machine.

Read and understand the basic precautions listed in the safety section before operating or performing lubrication, maintenance and repair on this product.

Operation

This manual includes sections which explain system gauges and switches, resurfacer and attachment controls and basic operating techniques.

Operating techniques outlined in this publication are basic. Skill and techniques develop as the operator gains knowledge of the Zamboni ice resurfacer and its capabilities.

1.

General Safety Practices

Safety



This symbol warns of possible personal injury.

The safety rules and regulations in this section are representative of some, but not all, rules and regulations. The Canadian Centre for Occupational Health and Safety (CCOHS) is paraphrased without representation that the CCOHS rules and regulations have been reproduced verbatim. Please refer to your provincial Health and Safety regulatory body for a complete list of rules and regulations. Regulations vary from province to province as well as country to country; operate your Zamboni ice resurfacer in accordance with local regulations.

Important Safety Notice

Periodic and proper operation, lubrication and maintenance is important for the safety and reliability of your Zamboni machine. The manual outlines recommended procedures, some of which require the use of special tools or work methods.

DANGER!



Improper operation, lubrication and maintenance of this machine is dangerous and could result in injury or death.

READ AND UNDERSTAND ALL SAFETY PRECAUTIONS AND WARNINGS BEFORE OPERATING, LUBRICATING OR PERFORMING MAINTENANCE ON THIS MACHINE.

Basic safety precautions are outlined in the SAFETY section of this manual and in the description of operations where hazards exist. Warning labels have also been put on the machine to provide instructions and to identify specific hazards which could cause bodily injury or death to you or other persons.

It is the responsibility of the owner to perform a job safety analysis of all hazards associated with the total environment in which the Zamboni machine will operate and to train all personnel annually.

The warnings in the manual and on the machine are identified by the symbols:

DANGER!



This safety symbol warns of possible personal injury or death.

WARNING!



This safety symbol warns of possible personal injury.

CAUTION!



This symbol warns of possible damage that may occur to the machine.

DANGER!



Operations that may cause product damage are identified on the product and this publication. Zamboni cannot anticipate every possible circumstance that might involve potential hazard.

The warnings in this manual and on the machine are therefore not all-inclusive. If a procedure, tool or work method not specifically recommended by Zamboni is used, a competent person must be consulted to assure the safety of you and others and that the machine will not be damaged or made unsafe by the procedures you choose. You should also ensure that the product will not be damaged or made unsafe by the operation, lubrication, maintenance or repair procedure you choose. When in doubt, call our factory for free consultation and advice.

DANGER!



Do not operate or perform maintenance on this machine unless you have read and understand the instructions in THIS MANUAL. Improper machine operation and maintenance is dangerous and could result in injury or death. Contact Zamboni for a free replacement manual. Proper operation and maintenance is your responsibility.

Accidents involving product operation, maintenance and repair are caused by failure to observe safety rules or precautions. A person must be alerted to potential hazards. This person should also have the necessary training, skills and tools to perform safely.

The specifications are subject to change at any time. These changes may affect the service given to the product. Please obtain the most current information from your local distributor or the Zamboni factory before starting any job.

For a list of the most current publications please contact: Zamboni Company Ltd., 38 Morton Avenue East, P.O. Box 1388, Brantford, Ontario, Canada N3T 5T6 (519) 758-5000

Warning Signs and Labels

There are several specific safety signs on your Zamboni machine.

Make sure that you can read and understand all of the safety signs. Clean or replace them if you cannot read the words or see the pictures. When cleaning the labels use a cloth, water and soap. Do not use solvent, gasoline, etc.

You must replace a label if it is damaged, missing or cannot be read. If a label is on a part that is replaced, make sure a new label is installed on the replaced part.

Contact Zamboni Company Ltd. for new safety labels free of charge.

DANGER!



Only trained and authorized personnel may operate this machine. For safe operation, read and follow the manual furnished with this machine and observe the following warnings:

- Keep all guards in place. Do not operate the Zamboni machine if any guard is damaged or missing.
- Check all controls and warning devices for proper operation.
- Insure directional control is in neutral before “ON-OFF” switch is turned on.
- Start, turn and brake smoothly. Slow down for turns, slippery or uneven surfaces. Use extreme caution when turning or on inclines.
- Watch out for pedestrians and obstructions, check overhead clearances. Use care when backing up.
- Do not permit riders on the Zamboni machine at any time.
- Do not permit skaters or people on the ice when the machine is operating on the ice.

- Do not allow anyone to stand or pass under the elevated portion of any machine.
- Be sure operating surface can safely support machine.
- Observe safety rules when handling fuel on engine powered machines and when changing or charging batteries for electric machines.
- Never cover, hide or remove any safety label.
- Never stand on the ice making water tank. Damage to the tank and personal injury may occur.

Crushing or Cutting Prevention

- Support equipment and attachments properly when working beneath them.
- Do not depend on hydraulic cylinders to hold up. Any attachment can fail if a control is moved or if a hydraulic line breaks.
- Never attempt adjustments while the machine is moving or is running unless otherwise specified.
- Where there are mechanical linkages, the clearance in the linkage area will increase or decrease with movement of the assembly.
- Stay clear of all rotating and moving parts.
- Keep objects away from moving fan blades. They will throw or cut any object or tool that falls or is pushed into them.
- Retainer pins, when struck with force, can fly out and injure nearby persons. Make sure the area is clear of people when striking retainer pins.
- Wear protective glasses when striking a retainer pin to avoid injury to your eyes.
- Chips or other debris can fly off objects when struck. Make sure no one can be injured by flying debris before striking any object.
- Keep all guards in place and good repair.

Fire or Explosion Prevention

Fuels (Gasoline, Propane - LPG, Compressed Natural Gas - CNG)

DANGER!



Gasoline, LPG , CNG and their vapors are extremely flammable. LPG and CNG vapors reduce oxygen available for breathing and may cause suffocation. Gasoline vapors are extremely flammable and cause flash fires.

- Physical damage such as dents, scrapes or gouges may materially weaken the structure of the tank and render it unsafe for use.
- All LP gas containers should be regularly inspected before recharging and examined again before reuse, for the following defects or damage.
- Keep container valve closed and plugged when not in use.
- Never use a match or open flame to check for leaks, use a soap solution.
- Always close the LPG tank service valve, remove the tank(s) and store them in an outside, locked cage when the Zamboni machine is parked for the night or for any extended length of time.
- Never smoke when filling, installing or changing the fuel tanks on the machine.
- Inspect for deterioration, damage or loss of flexible seals in the fill or servicng connection.
- Do not overfill LP gas containers.
- Keep all sources of open flame, spark or other ignition such as stoves, furnaces, water heaters or any appliance using a pilot light away from fuel storage, fuel use and refueling areas. Under well ventilated conditions, a minimum distance of 20 feet from all ignition sources is recommended.
- Do not operate the machine if an odor of gasoline or LPG or CNG is present or if other explosive conditions exist.
- Refuel outdoors or in a well ventilated area.
- DO NOT refuel when the engine is running or hot. Allow engine and engine exhaust system and catalytic convertor to cool for a minimum of five minutes before refueling.
- Never fill the gasoline tank completely full or to a point where the gasoline is overflowing. Always leave a small portion of the gas tank and the entire fill tube empty. This will allow for the expansion and movement of the gasoline and will minimize sloshing and potentially spilling the gasoline when the machine moves.
- If gasoline is spilled, avoid creating any source of ignition until the gasoline has been completely removed.
- LPG is heavier that air and spreads rapidly. Stop all leaks quickly. Shut off the source of gas.
- If a fire starts, evacuate the area and notify the fire department.

- For a fire in the engine compartment, shut off engine. Shut off the gas flow at the tank valves for LPG and CNG powered machines before trying to extinguish.
- Always have a properly maintained fire extinguisher located in the Zamboni machine room.
- Fire extinguishing media must be dry chemical, foam, or CO_2 for usage on large fires.

Propane (LPG) Tank Storage and Handling

- Avoid skin contact with liquid because of possibility of freeze burn. Use propane resistant gloves when working with the propane tank connections. Always use full face mask and wear a long sleeve shirt or jacket.
- The storage and handling of LP gas and tanks should be in accordance with the CAN/CSA -B149.2 Propane Storage and Handling Code, 2005 Edition.
- The machine should be refueled in accordance with the CAN/CSA -B149.2 Propane Storage and Handling Code, 2005 Edition.
- Only properly trained and designated persons should charge or exchange the LP gas tanks.
- The engine should be stopped and no persons should be on the Zamboni machine during refueling.
- Always install tanks and make tank connections outdoors or in a well ventilated area away from heat, ignition sources and open flames.
- NO SMOKING ALLOWED during these operations!
- Store tanks in an authorized location (outside, detached storage is required) with adequate ventilation.
- Store tanks in an upright position, away from heat and ignition sources.
- Reasonable care should be exercised in handling the LP gas containers to avoid damage. Do not drop, throw, roll or drag LP gas containers or any associated parts of the containers or fuel system.
- The careless handling of LP gas containers can result in a serious accident. Extreme care should be exercised when transporting containers so that they are not accidently dropped or physically damaged. When it is necessary to move more than one container at one time, a proper carrying device should be provided and used.

- Physical damage such as dents, scrapes or gouges may materially weaken the structure of the tank and render it unsafe for use.
- All LP gas containers should be regularly inspected before recharging and examined again before reuse for the following defects or damage:
    1. Leaks, dents, scrapes, gouges or corrosion of the pressure vessel, with emphasis on the bottom of the cylinders.
    2. Damage to the various valves and liquid level guage.
    3. Debris in the relief valve.
    4. Indication of leakage at valves and/or threaded connection.
- Keep container valve closed and plugged when not in use.
- Never use a match or open flame to check for leaks, use a soap solution.
- Always close the LPG tank service valve, remove the tank(s) and store them in an outside, locked cage when the Zamboni machine is parked for the night or for any extended length of time.

CNG Fuel System

The installation of distribution, storage and dispensing (charging) systems must be done in accordance with the ANSI/IAS NGV4.1-1999 / CSA 12.5-M99 (R04) and any other applicable regulations. Your local gas utility company must be advised of your installation.

The machines's CNG cylinders must be charged in accordance with the ANSI/IAS NGV4.1-1999 / CSA 12.5-M99 (R04).

Only properly trained and designated persons should charge the CNG fuel cylinders on the machine.

- Never smoke when charging the CNG fuel cylinders on the machine.
- Keep all sources of open flame, spark or other ignition such as stoves, furnaces, water heaters or any appliance using a pilot light away from fuel storage, fuel use and refueling areas. Under well ventilated conditions, a minimum distance of 20 feet from all possible ignition sources is recommended.
- Never charge the cylinders in excess of their maximum allowable service pressure at normal temperature. Department of Transportation and Transport Canada containers shall be charged in accordance with DOT and TC regulations.

Oils

- Hydraulic oil and its vapors are flammable and will ignite if they contact a hot surface such as an exhaust pipe or manifold. A flash fire can occur.
- Hot oil and components can cause personal injury. Do not allow hot oil or components to contact the skin.
- At operating temperature, the hydraulic tank is hot and can be under pressure.
- Remove the hydraulic tank filler cap only after the machine has been stopped and the filler cap is cool enough to remove with your bare hand.
- Remove the hydraulic tank filler cap slowly to relieve pressure.

Batteries

- Battery fumes may cause fire or explosion. Battery acid can cause burns.
- Only trained and designated personnel should recharge or exchange batteries.
- Service, exchange and handle batteries only in authorized areas where proper safety and ventilation facilities are provided (i.e. eye wash stand, showers, etc.).
- Do not smoke, or expose battery to sparks or flame when checking, charging or servicing battery. Keep chains and metallic tools away from the top of the battery.
- Highly explosive gases are present and are especially hazardous toward the end of the charging period as the battery approaches a full charge condition.
- Cover the top of the battery with plywood or other insulating material before removing it from the machine.
- Do not add acid to a “wet” battery, add distilled water only.
- When preparing a dry charged battery for service, follow the battery manufacturer’s instructions.
- Do not smoke in areas where batteries are charged or where flammable materials are stored.
- Clean and tighten all electrical connections. Check daily for loose or frayed electrical wires. Have all loose or frayed electrical wires tightened, repaired or replaced before operating the Zamboni ice resurfacer.

Lubricants

Most lubricants are flammable.

- Keep all lubricants stored in properly marked containers and away from unauthorized persons.
- Store all oily rags or other flammable material in an approved container, in a safe place.
- Do not weld or flame cut on pipes or tubes that contain flammable fluids. Clean them thoroughly with non-flammable solvent before welding or flame cutting on them.
- Remove all flammable materials such as oil and other debris before they accumulate on the Zamboni machine.
- Do not expose the ice resurfacer to flames, burning, etc. if at all possible.
- Have a fire extinguisher available and know how to use it. Inspect and have it serviced as recommended on its instruction plate.

Lines, Tubes and Hoses

- Do not bend or strike high pressure lines. Do not install bent or damaged lines, tubes or hoses.
- Repair any loose or damaged oil lines, tubes and hoses. Leaks can cause fires. Contact the Zamboni Company or your Zamboni machine dealer for repair or replacement.
- Check lines, tubes and hoses carefully. Do not use your bare hand to check for leaks. Use a board or cardboard to check for leaks. Tighten all connections to the recommended torque. Replace if any of the following conditions are found:
    - End fittings damaged or leaking.
    - Outer covering chafed or cut and wire reinforcing exposed.
    - Outer covering ballooning locally.
    - Evidence of kinking or crushing of the flexible part of hose.
- Make sure that all clamps, guards and heat shields are installed correctly to prevent vibration, rubbing against other parts and excessive heat during operation.

Safe Operation

Mounting and Dismounting

- Mount and dismount the ice resurfacer only where steps and/or handholds are provided.
- Use both hands and face the Zamboni machine when mounting and dismounting. Use extra care when wet or slippery conditions exist.
- Never get on or off a moving Zamboni machine. Never jump off the machine.
- Do not try to climb on or off the ice resurfacer when carrying tools or supplies.

Before Operating the Zamboni Ice Resurfacer

- Operate the machine only from the operator's station.
- Adjust the seat so that full pedal travel can be obtained with the operator's back against the seat back.
- The Zamboni machine is equipped with a lighting system, make sure all lights are working properly.
- Make sure no one is working on, underneath or close to the ice resurfacer before turning on the key or beginning to move the machine. Make sure the area is free of personnel.
- Do not move the machine or move any of the controls if there is a “DO NOT OPERATE” or similar warning tag attached to the start switch or controls.
- Clear all personnel from the machine and the area.
- Clear all obstacles from the path of the ice resurfacer. Beware of hazards such as curbs, blocks, posts, wire, cans, etc.
- Make sure the horn and all other warning devices are working properly.
- Check for proper operation of all controls.

Operation

- Obey all traffic rules and warning signs.
- Always observe floor load limits and overhead clearances.
- When turning the ice resurfacer, slow down to safe speed.

Parking

- Park the Zamboni machine in authorized areas only.
- Remove foot from the acceleration foot pedal to place the machine in NEUTRAL.
- Lower the conditioner onto blocks.
- Turn the ignition switch OFF and remove the key.
- If possible, park the machine on a flat and level surface. If the machine is parked on any kind of grade, the wheels must be blocked with wheel chocks to prevent the machine from rolling away after it has been parked.
- Apply the parking brake if the parking brake option was purchased with the machine.
- Store in well ventilated area.

Maintenance

Walk-Around Inspection

For maintenance and operator personnel safety and maximum service life of the Zamboni machine, make a thorough walk-around inspection when doing lubrication and maintenance work. Look around and under for such items as loose or missing bolts, trash or dirt build-up, oil leaks and cut or gouged tires.

DANGER!



Improper performance of lubrication or maintenance procedures is dangerous and could result in injury or death. Read and understand all OPERATION & MAINTENANCE instructions before performing any lubrication or maintenance.

DANGER!



Do not operate this machine unless you have read and understand the instructions in the OPERATOR'S manual. Improper machine operation is dangerous and could result in injury or death.

The service person or mechanic may be unfamiliar with many of the systems on this machine. This makes it important to use caution when performing service work. A knowledge of the system and/or components is important before the removal or disassembly of any component.

The following is a list of basic precautions that should always be observed:

- Read and understand all warning plates and decals on the machine before operating, lubricating or repairing this product.
- The vertical and horizontal augers will continue to rotate for a short period of time after the auger control valves have been shut off. Always allow for all auger rotation to stop before working near or on the augers. Never work near or on the augers when the engine is operating.
- Always wear protective glasses and protective shoes when working around machines. In particular, wear protective glasses when pounding on any part of the machine or its attachments with a hammer or sledge. Use welder's gloves, hood/goggles, apron and other protective clothing appropriate to the welding job being performed. Do not wear loose, fitting or torn clothing. Remove all rings from fingers when working on machinery.
- Do not work on any machine that is supported only by jacks or a hoist. Always use blocks or jack stands to support the machine before performing any disassembly.

- Lower the conditioner or other implements to the ground before performing any work on the machine.
- Use steps and grab handles (if applicable) when mounting or dismounting a machine. Clean any mud or debris from steps, walkways or work platforms before using. Always face machine when using steps, ladders and walkways. When it is not possible to use the designated access system, provide ladders, scaffolds or work platforms to perform safe repair operations.
- Use proper lifting procedures when removing components. To avoid back injury, use a hoist when lifting components which weigh 50 lbs. (23 kg.) or more. Make sure all chains, hooks, slings, etc. are in good condition and are of the correct capacity. Be sure hooks are positioned correctly. Lifting eyes are not to be side loaded during a lifting operation
- To avoid burns, be alert for hot parts on machines which have just been stopped and hot fluids in lines, tubes and compartments.
- Be careful when removing cover plates. Gradually back off the last two bolts or nuts located at opposite ends of the cover or device and pry cover loose to relieve any spring or other pressure, before removing the last two bolts or nuts completely.
- Be careful when removing filler caps, breathers and plugs on the machine. Hold a rag over the cap or plug to prevent being sprayed or splashed by liquids under pressure. The danger is even greater if the machine has just been stopped because fluids can be hot.
- Always use tools that are in good condition and be sure you understand how to use them before performing any service work.
- Reinstall all fasteners with same part number. Do not use a lesser quality fastener if replacements are necessary. Do not mix metric fasteners with standard nuts and bolts.
- If possible, make all repairs with the machine parked on a level, hard surface. Block machine so it does not roll while working on or under machine.
- Repairs which require welding should be performed only with the benefit of the appropriate information and by personnel adequately trained and knowledgeable in welding procedures.
- Determine type of metal being welded and select correct welding procedure and electrodes, rods or wire to provide a weld metal strength equivalent at least to that of parent metal.

- Do not damage wiring during removal operations. Reinstall the wiring so it is not damaged nor will it become damaged in operation by contacting sharp corners, or by rubbing against some object or hot surface. Do not connect wiring to a line containing fluid.
- Be sure all protective devices including guards and shields are properly installed and functioning correctly before starting a repair. If a guard or shield must be removed to perform the repair work, use extra caution.
- Loose or damaged fuel, lubricant and hydraulic lines, tubes and hoses can cause fires.
- Do not bend or strike high pressure lines or install ones which have been bent or damaged. Inspect lines, tubes and hoses carefully. Do not check for leaks with your hands. Pinhole (very small) leaks can result in a high velocity oil stream that will be invisible close to the hose. This oil can penetrate the skin and cause personal injury. Use cardboard or paper to locate pinhole leaks.
- Tighten connections to the correct torque. Make sure all heat shields, clamps, and guards are installed correctly to avoid excessive heat, vibration or rubbing against other parts during operation. Relieve all pressure in air, oil or water systems before any lines, fittings or related items are disconnected or removed. Always make sure all raised components are blocked correctly and be alert for possible pressure when disconnecting any device from a system that utilizes pressure.
- Do not operate a machine if any rotating part is damaged or contacts any other part during operation. Any high speed rotating component that has been damaged or altered should be checked for balance before reusing.
- Caution should be used to avoid breathing dust that may be generated when handling components containing asbestos fibers. If this dust is inhaled, it can be hazardous to your health. Components in Zamboni machine products that may contain asbestos fibers are brake pads, brake band and lining assemblies, and some gaskets. The asbestos used in these components is usually bound in a resin or sealed in some way. Normal handling is not hazardous as long as airborne dust which contains asbestos is not generated.
- If dust which may contain asbestos is present, there are several common sense guidelines that should be followed:
    - Never use compressed air for cleaning.
    - Avoid brushing or grinding of asbestos containing materials.
    - For clean up, use wet methods or a vacuum equipped with a high efficiency particulate air (HEPA) filter.

- Use exhaust ventilation on permanent machining jobs.
- Wear an approved respirator if there is no other way to control the dust.
- Comply with applicable rules and regulations for the work place (for example in the U.S.A., OSHA requirements as set forth in a 29 CFR 1910.1001)
- Follow environmental rules and regulations for disposal of asbestos.
- Avoid area where asbestos particles may be in the air.

Battery Operation and Maintenance

WARNING!



Hydrogen and oxygen gases are given off as battery plates get near a full charge condition. These gases are highly explosive and caution must be taken to make sure no arc, spark, or flame comes into contact with the generated gases. The possibility of explosion is always present with exposure of the battery to sparks or flame, but is especially hazardous at the end of the charging period.

WARNING!



When using air pressure for cleaning purposes, wear a protective face shield and protective clothing. Maximum air pressure must be below 30 psi (207 kPa).

Do not smoke near batteries that are being stored or when checking the electrolyte level.

Electrolyte is an acid solution and can cause personal injury. Avoid contact with skin and eyes.

Maximum life and performance of batteries is dependent on the operator, battery charging, maintenance and service.

Most dirt and dust picked up by the battery can usually be blown off with low pressure compressed air.

However, if cells are overfilled and electrolyte collects on the covers, the top of the battery will stay wet.

If necessary, clean the top of the battery with a solution of baking soda and hot water.

CAUTION!



Vent caps must be tight to prevent soda solution from entering battery cell.

To make the solution, add 1 lb (0.5 kg) of baking soda to 1 gallon (4 liters) of water. Use a brush having flexible bristles. Apply the soda solution to the top of the battery until the cleaning action of the soda stops.

After cleaning action has stopped, rinse batteries thoroughly with water. Dry the batteries with air pressure.

Cold Storage Applications

When a Zamboni machine is operated in very cold applications at temperatures as low as -4 degrees Fahrenheit (-20 degrees Celsius), the battery capacity is decreased. Operation at cold temperatures can also cause mechanical failures, short circuits and too much wear due to the formation of ice crystals.

The direct cause of these problems is the extreme changes in temperature in combination with humidity in the air which can result in condensation.

There is a reduction in battery capacity in very cold applications. For this reason, it is important to:

- Be sure the battery is completely charged at the start of each work cycle.
- If possible, keep the Zamboni machine in a warm storage area when it is not in use.

2.

General Operating and Maintenance Information

Important :

The engine, drive train and differentials on the ice resurfacer are all new components and special care must be given to them during their "break-in" period.

Service schedules for machine lubrication and preventative maintenance have been established on the basis of hourly, daily, weekly and monthly intervals (see 526 Service Manual). Along with performing regular machine lubrication and preventative maintenance according to these schedules, it is advised that you have the ice resurfacer and its components completely checked over periodically by a competent mechanic.

Study should be given to the Mitsubishi Operating and Service manuals which are furnished with your ice resurfacer. It would be beneficial to have your service person inspect the engine and drive system so that they may become familiar with the operation for service and repair. It is also recommended that the new owner contact the local Zamboni Authorized Dealer to acquaint themselves with the service policy.

Note Regarding the Mitsubishi Engine:

The engine is equipped with a low oil pressure warning light and an alternator warning light mounted on the instrument panel. The engine also has a "malfunction indicator light (MIL)" which warns the operator about engine conditions which may require service. For more information on the "malfunction indicator light", see Chapter 4 and Chapter 6.

Grade of Engine Oil

Use only engine oils displaying the American Petroleum Institute (API) "Starburst" Certification Mark 'FOR GASOLINE ENGINES' and a rating of at least SJ on the container. Oil capacity is 5.7L (5 quarts) with filter.

Fuel Recommendation

This engine is designed to operate on unleaded 87 or 89 octane gasoline. The engine, with the proper fuel equipment, can also be operated on dry fuel such as LPG and NG.

NOTE: It is highly recommended that a fuel stabilizer additive be used for any length of storage. It is imperative in any application where the fuel will not be consumed within thirty days.

LPG fuel specifications must meet or exceed EN 589 (European) HD-5 (Canada/USA). NG fuel specification must meet or exceed 38.7 MJ/m^3 (European) 39.0 MJ/m^3 (Canada/USA) (1050BTU/ft^3).

CAUTION!



Use of fuels rated lower than the grades specified above may cause persistent, heavy spark knock, which can lead to engine damage. If your engine knocks heavily when using unleaded gasoline with 87 octane or higher, or if you hear continuous spark knock while maintaining constant operating speeds, consult your distributor or another qualified technician.

Your efforts spent in properly maintaining the ice resurfacer and its various components will be well repaid by longer wear and good service.

Starting the Engine

To start the engine:

- Make sure the acceleration foot pedal is in the Neutral position.
- Make sure the auger control levers are Off/Neutral.
- Turn ignition key to start.
- The engine will start and run at approximately 1150 RPM. Wait 15 seconds before increasing the engine speed for resurfacing. This warm up period will allow the engine and catalytic convertor to achieve the correct operating temperatures. After the correct warm up period the engine will be operating in “closed loop” and the catalytic convertor will be more efficient in reducing engine exhaust emissions. Be sure that the engine emissions are exhausted to the outside, or the room is properly ventilated.
- Be sure snow tank is completely down and the conditioner is raised.

Stopping the Engine

To stop the engine:

- Return the acceleration foot pedal to Neutral. Set the parking brake, if so equipped.
- Return the auger control levers to Off/Neutral.
- After the snow tank has been emptied, lower the conditioner onto blocks.
- Lower the engine RPM to idle by using the engine speed switch. **Let the engine idle for 30 to 60 seconds before turning the ignition off** to allow the engine and exhaust system to cool down.
- If the snow tank is up, install the safety stand to the correct position (see Fig. 5.6).
- Turn off the ignition key switch and remove the ignition key to store in a secure location.

Machine Not in Use

When the machine is not in use:

- Park the machine in authorized areas only.
- Lower the conditioner onto blocks. The conditioner weighs approximately 1000 pounds. This weight should not hang on the lift bar nor be supported by the hydraulic system during storage. Lowering the conditioner onto the blocks must be done cautiously as putting hydraulic down pressure on the blocks (in excess of the weight of the conditioner) can stretch the down pressure leaf springs. Wood blocks should be placed under both conditioner runners.
- Return the auger control levers to Off/Neutral. Set the parking brake, if so equipped.
- Remove and securely store the ignition key.
- If possible, park the machine on a flat and level surface. If the machine is parked on any kind of grade, the wheels must be blocked with wheel chocks to prevent the machine from rolling away after it has been parked.

Travel on Highways

The machine is not designed for highway use. Fast driving on the highway and bouncing the machine over bumps can seriously damage the machine. Travel over extended distances should be avoided, if possible. If such travel is required:

- Check local authorities to determine if travelling on highways is permitted.
- Vehicle escort in front of and behind ice resurfacer is recommended.
- Exercise Caution. Obey all traffic signs, signals and laws.
- Make sure that the conditioner is completely raised before driving.
- Do not exceed engine speed of 2500 RPM.
- Do not tow the machine for long distances. Transmission damage will occur.

3.

Safe Operation

Your Zamboni ice resurfacer is the result of many years of experience in the ice resurfacing field. The engineering and safety features that have gone into your Zamboni machine will be enhanced by you, the safe operator...

...who knows his machine and all of the controls,

...who maintains his machine properly,

...who uses his driving skills wisely and operates the machine with safety in mind.

Because SAFE DRIVING is important to YOU, we urge you to thoroughly acquaint yourself with the resurfacer and the instructions contained in the Operation Manual. Even the safest machine can be operated improperly so certain precautions are mentioned in this manual.

Among these precautions are:

- Be sure that all guards originally installed on the machine are properly in place and in good working order.
- Make sure the hydrostatic transmission is in neutral before starting the engine. Always look completely around the machine prior to moving forward or reverse to insure that no one is in the path of your direction of travel.
- The ice resurfacer is to be operated by one, and only one person. **No riders!**
- Keep everyone off the ice while the resurfacer is operating. Always drive defensively. Expect the unexpected.
- Keep hands and feet away from conveyors, belts, chains and other moving parts while the engine is running.
- Allow the engine and catalytic convertor to achieve the correct operating temperatures ***before driving and resurfacing.***
- The Zamboni machine is an industrial machine designed to be used by trained and skilled operators. It is never to be used for recreational purposes, such as racing.

DANGER!



Turn off and remove the ignition key before adjusting, repairing, cleaning, or servicing the conveyors. The vertical and horizontal augers will continue to rotate for a short period of time after the auger control valves have been shut off. Always allow for all auger rotation to stop before working near or on the augers. NEVER work near or on the augers when the engine is operating.

If any part of the conveyor system becomes plugged or clogged, or appears plugged or clogged, the conveyor must be flushed out with water. Do not wear gloves or jewelry when performing these functions.

Flushing the augers out with water must be done only after the machine has been removed from the ice, the auger control valves have been returned to their "neutral (off)" positions, and the ignition has been shut off.

Under no circumstances should the operator attempt to clean or clear the conveyor system or parts around it while the machine is on the ice or the power is on. To do so, exposes the operator and others to serious potential injury or death!

WARNING!



Keep hands, feet and body out from under the snow tank, conditioner and other raised parts unless they are blocked securely. Use the safety stand under the snow tank.

Be extremely careful when handling the shaving knife. Unless handled with care, the blade can cause injury whether it is sharp or dull.

WARNING!



If the machine is parked on any kind of grade, the wheels must be blocked with wheel chocks to prevent the machine from rolling away after it has been parked.

DANGER!



Internal combustion engines produce exhaust emissions that contain dangerous gases, including carbon monoxide (CO) and nitrogen dioxide (NO_2). These gases can cause serious injury or death.

The rink MUST be adequately ventilated during every operation of the ice resurfacer. Also the resurfacer must be kept properly maintained and serviced at all times and the engine timing should always be properly set. Each person who operates and maintains the resurfacer should be so instructed.

Please keep in mind that adequate ventilation, monitoring the facility's indoor air quality (for CO and NO_2 levels) and the condition of the machine is, at all times, the complete responsibility of the rink.

- Use extreme caution when handling fuel and store only in approved containers.
- Annual training of the operator is required.
- All new operators must be trained before operating the machine for the first time.
- Warning labels should be followed. Do not operate if the labels are missing or cannot be read.

DANGER!



This machine has an electronic engine control unit (ECU). If the ECU detects that the engine is operating outside of it's desired parameters, the "check engine" or malfunction indicator light (MIL) will be turned on. Do not operate the machine with the "check engine" light illuminated for a continuous period. Have the machine serviced as soon as possible.

4.

Ice Rink Air Quality

In order for a rink to have the best air quality possible, the Zamboni machine must be operated properly in a rink that:

- Is adequately ventilated.
- Performs daily air quality monitoring/measuring.
- Knows which air quality standards or guidelines apply to them.
- Knows which items on the Zamboni machine affect air quality and how and when these items should be measured and serviced.
- Allows the engine and catalytic converter to properly warm up before operating the machine on the ice.

Adequate Ventilation

Adequate ventilation is the amount of ventilation required to ensure that the air is healthy to breathe and meets all applicable health and safety standard or guidelines.

Air Quality Monitoring

Regular air quality monitoring is a program that each rink sets up to ensure that their air quality is healthy and meets the standards.

Air quality standards or guidelines define the amount of Carbon Monoxide gas (CO) and Nitrogen Dioxide gas (NO_2) that can exist in the air inside the ice rink. These gases are exhaust emissions from the combustion of fossil-based fuels such as gasoline, LPG and CNG.

Ice rink air quality can also be affected by many other factors. These include:

- Ice edgers powered by internal combustion engines.
- Unvented propane heaters, furnaces, or boilers.
- CO_2 tanks.
- Fuels with impurities.
- The ice refrigeration system and it's gases (ammonia or "Freon").
- Cigarette smoke.
- Restaurant and/or vendor equipment.
- Improper storage of paints, chemical, solvents, etc.

The arena's ventilation system can also dramatically affect the indoor air quality by:

- Not being properly operated.
- Not being properly maintained.
- Not being properly designed such as:
    - Inadequate size or performance.
    - Intake air systems drawing in air from other polluted areas, such as a vehicle parking lot or garage.

Even cars, buses or trucks idling next the arena doors or ventilation system intakes can have a dramatic affect upon the indoor air quality.

Engine Emission System Components

This machine is equipped with an engine system that is designed to reduce emissions to the point of compliance with industry standards. In order to comply with these standards, the machine is equipped with the following items:

- Electronic Control Unit (ECU).
- Electronic Fuel Injection (EFI - Gasoline Only).
- Electronic governor.
- Three-way catalytic converter.
- Oxygen sensor.
- MAP sensor.
- Engine coolant temperature sensor.
- Ambient air temperature sensor.
- On-board diagnostic system.
- Malfunction indicator light (MIL).

All of these items are used to insure that the engine operates properly and meets the emission level standards. However, exhaust emissions are still being produced by the engine and while the level of the harmful pollutants are reduced, the air quality in the rink must still be monitored to make sure that the air is healthy.

Engine Control Unit (ECU)

The ECU works to control the engine's air-fuel ratio (AFR) to the ideal levels based on the engine's operating conditions. The ECU monitors the engine for speed, load, air and coolant temperature and then adjusts the AFR via the electronic fuel injection to produce the ideal mixture. If the AFR is near the ideal ratio (stoichiometric), then the catalytic converter will operate at it's maximum efficiency thus reducing the emission levels of the harmful exhaust gases.

Typically, it takes a few minutes for the engine and catalytic converter to achieve proper operating temperatures and conditions. Resurface only when the engine emission system is up to normal operating temperature.

If the ECU detects conditions where it cannot control the AFR, it will then turn on the MIL, thus alerting the operator. If the ECU detects that a portion of the engine equipment, which it monitors, is malfunctioning, it will turn on the MIL. DO NOT OPERATE the machine with the malfunction indicator light on continuously.

DANGER!



If the check engine light turns on during normal operation of the machine, the machine must be serviced immediately!

Open Loop versus Closed Loop Operation

When the engine is first started, it is operating in "***open loop***" condition. This means that the ECU is controlling the AFR via pre programmed levels. When the engine achieves a certain operating temperature, the ECU then goes into the "***closed loop***" mode. This means that the ECU is controlling the AFR based on the engine operating conditions and is monitoring the signal from the oxygen sensor to determine if changes to the AFR need to be made.

With the precise closed loop control of the AFR, the catalytic converter will be the most effective in reducing the exhaust emission levels. The lowest levels of harmful emissions are produced when the machine is operating in the closed loop mode. For this reason, start the engine and warm it up in accordance with the starting instructions described on page 2-2. ONLY operate the machine on the ice when the engine emission system is up to normal operating temperature.

Oxygen Sensor

The oxygen sensor is a critical component used by the engine controller. Correct oxygen sensor operation should be verified at each engine service or tune-up (every 1000 hours). The oxygen sensor must be replaced every **2000 hours** of operation.

Catalytic Converter

Your Zamboni ice resurfacer is equipped with a "three-way" catalytic convertor. The catalytic convertor is designed to reduce the amount of certain exhaust gases. A three way catalytic reduces CO, hydrocarbons and NO_X, (NO_2 is a member of the NO_X family).

Catalytic convertors are items that need regular service. A catalytic convertor may be damaged by incorrect engine timing or by an engine air-fuel ratio that is incorrect. Operating the machine with the check engine light on may reduce the performance of the catalytic convertor and may also shorten the life expectancy of the catalytic convertor.

A catalytic convertor has a **fixed** lifespan. It may lose efficiency and start to allow more pollution to pass through it without being treated.

DANGER!



If the check engine light turns on during normal operation of the machine, the machine must be serviced immediately!

Engine Ignition Timing

The engine ignition timing is one important factor that will affect the engine's production of exhaust emissions. The engine timing is factory set and is not designed to need adjustment.

A machine with advanced timing can produce high levels of NO_x and one with retarded timing can produce high levels of CO. Either of these conditions can adversely affect the catalytic convertor performance and life.

DO NOT ADJUST THE ENGINE TIMING!

Engine and State of Tune

The engine’s condition and state of tune refer to how the engine is operating and whether the operating variables of the engine have been adjusted to provide optimal performance.

An engine tune-up includes:

- Changing the spark plugs and noting their condition and appearance.
- Changing the distributor cap and rotor (if required).
- Changing the spark plug wires (if required).
- Inspecting all emission related components for correct function and for correct attachment to the engine.
- Noting and recording the condition of the malfunction indicator light.
- Noting and recording any fault codes displayed by the engine's ECU.
- Correcting any engine condition that produces a fault code or the display of the MIL.
- Recording work performed, dates of service and who performed the service.

Engine Emission Systems - Inspection and Maintenance

DANGER!



Internal combustion engines produce exhaust emissions that contain dangerous gases, including carbon monoxide (CO) and nitrogen dioxide (NO_2). These gases can cause serious injury or death.

The rink MUST be adequately ventilated during every operation of the ice resurfacer. Also the resurfacer must be kept properly maintained and serviced at all times and the engine timing should always be properly set. Each person who operates and maintains the resurfacer should be so instructed.

Please keep in mind that adequate ventilation, monitoring the facility's indoor air quality (for CO and NO_2 levels) and the condition of the machine is, at all times, the complete responsibility of the rink.

The safe and correct operation and maintenance of the resurfacer, the engine and the engine exhaust system includes:

- Measuring the facility's indoor air quality **daily** (for CO and NO_2 levels).
- Measuring and recording the engine emission levels every 300 hours. The emission levels must not have increased more than 20% from the previous readings.
- Adjusting engine components (when applicable) to ensure correct operation and optimum engine emission levels.
- Inspecting, adjusting (when applicable), maintaining and replacing engine and exhaust system components when required.
- Starting and warming up the engine properly so that the machine is in "closed loop" mode when on the ice.
- Never operating the machine with the check engine light on continuously.

If you do not have the tools, education or expertise to inspect, maintain, repair or replace these items, contact a reputable, licensed mechanic for assistance.

Engine Emission System - Inspection and Maintenance Chart

Serial No.: _____ **Fuel Type:** _____ **Catalytic Converter Type:** _____

Date	By	Hours	Engine		Cond.	After Catalytic Converter												
							HC	CO	CO_2	O_2	NO_X	Levels OK compared to previous testing?						
							PPM	%	%	%	PPM							
					Idle													
					Load													
					Idle													
					Load													
					Idle													
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Note: To load the engine, move the conditioner to fully lifted and hold for 15 seconds.

Recommended Maintenance Schedule											
GASOLINE AND LPG CERTIFIED ENGINE MAINTENANCE REQUIREMENTS											
	Install Date	Interval Hours									
		Daily	200	400	800	1000	1250	1500	1750	2000	5000
General Maintenance Section											
Visual Check for Fluid Leaks		X									
Check Engine Oil Level		X									
Check Coolant Level		X									
Change Engine Oil and Filter	Every 150 Hours or 120 Days of Operation										
Check LPG/Gas System for Leaks	Prior to any Service or Maintenance Activity										
Inspect Accessory Drive Belts						X					
Inspect Electrical System										X	
Inspect Vacuum Lines and Fittings										X	
Engine Coolant Section											
Clean Debris from Radiator Core	Every 100 Hours or 60 Days of Operation										
Change Coolant										X	
Inspect Coolant Hoses for Cracks, Swelling or Deterioration						X					
Engine Ignition System											
Inspect Battery Case for Damage						X					
Check all Electrical Connectors						X					
Replace Spark Plugs						X					
Base Engine System											
Replace Camshaft Belt											X
Replace Balance Shaft Belt											X
Inspect PCV System						X					
Fuel System Maintenance											
Replace Fuel Filter (Gas & LPG)				X							
Inspect Lock Off for Leaks & Closing										X	
Check LPG/Gas Regulator Pressure										X	
Leak Check LPG/Gas Fuel Lines										X	
Check/Drain EPR-LPR for Oil Buildup	Every 150 Hours or 120 Days of Operation										
Check Air Induction for Leaks										X	
Check Manifold for Vacuum Leaks										X	
Check Injector & Rail for Leaks										X	
Inspect Air Cleaner	Every 200 Hours, or Every 100 Hours in Dusty Environment										
Replace Filter Element	Every 400 Hours, or Every 200 Hours in Dusty Environment										
Engine Exhaust System											
Inspect Exhaust Manifold for Leaks										X	
Inspect Exhaust Piping for Leaks										X	
Inspect Catalyst Inlet and Outlet										X	
Check HEGO (O_2) Sensor Connections										X	
Change HEGO (O_2) Sensors										X	

5.

Ice Resurfacer Components

Front View

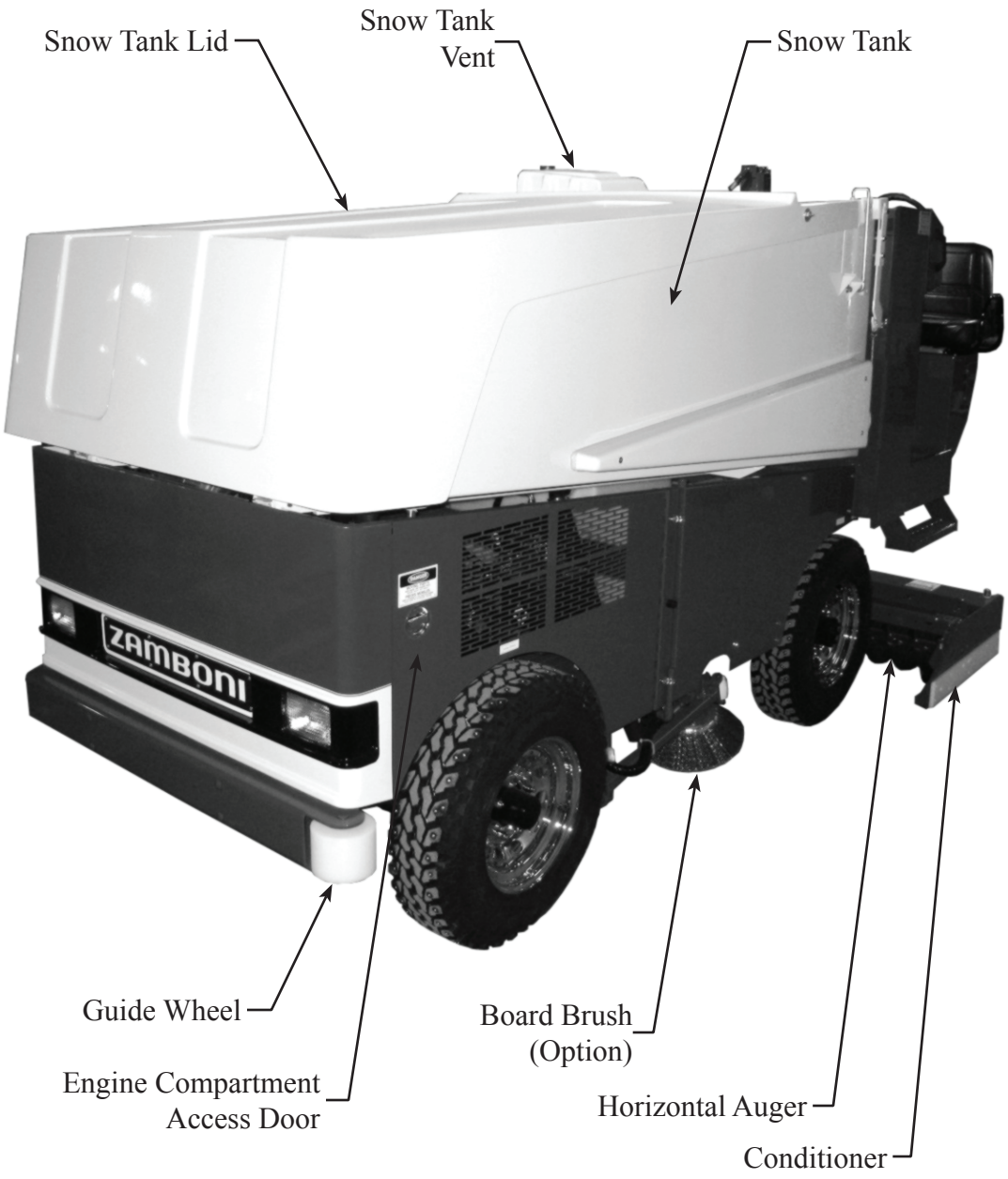


Figure 5.1 *Ice Resurfacer Components (front view)*

Rear View

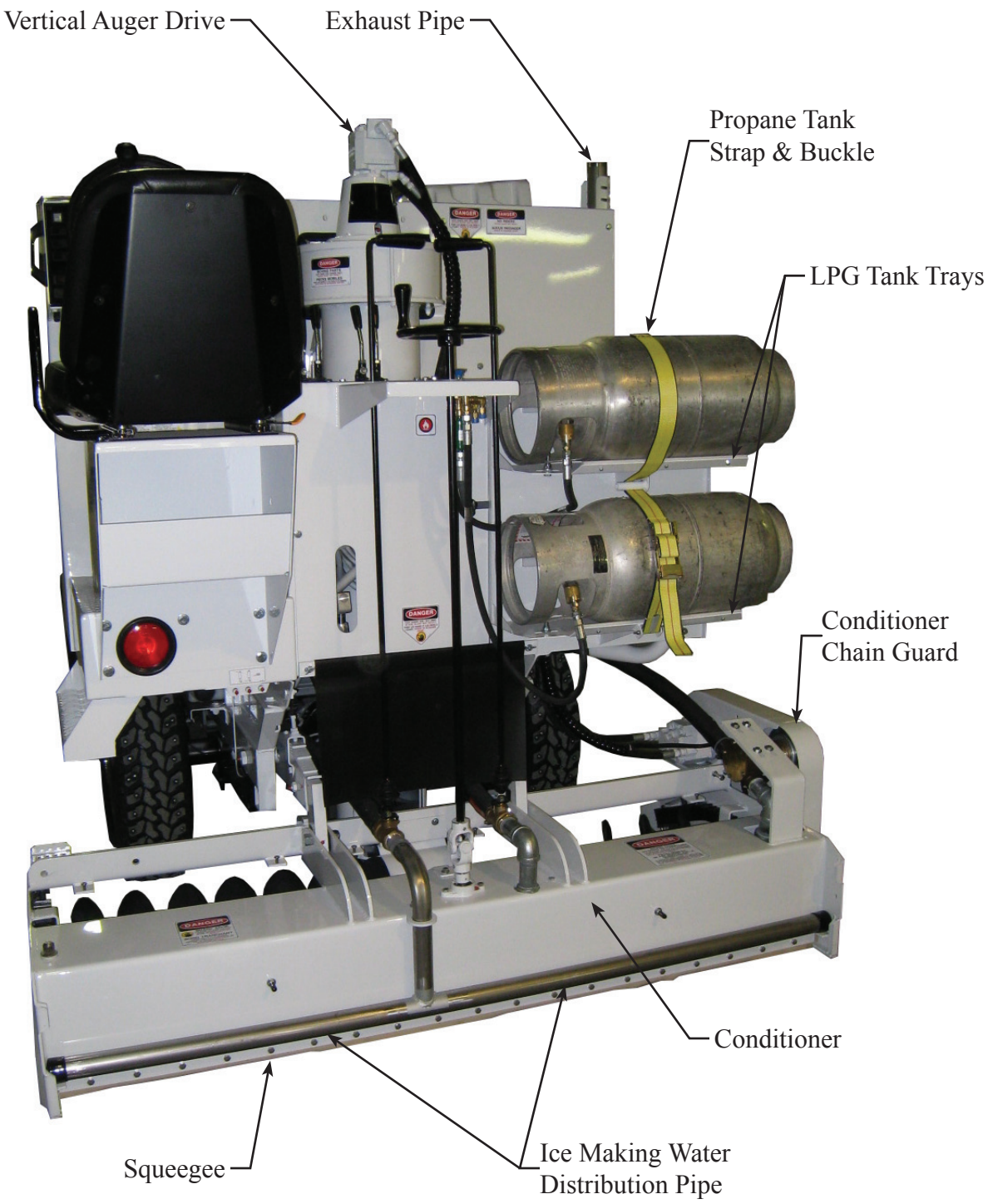


Figure 5.2 *Ice Resurfacer Components (Rear View)*

Operator's Platform

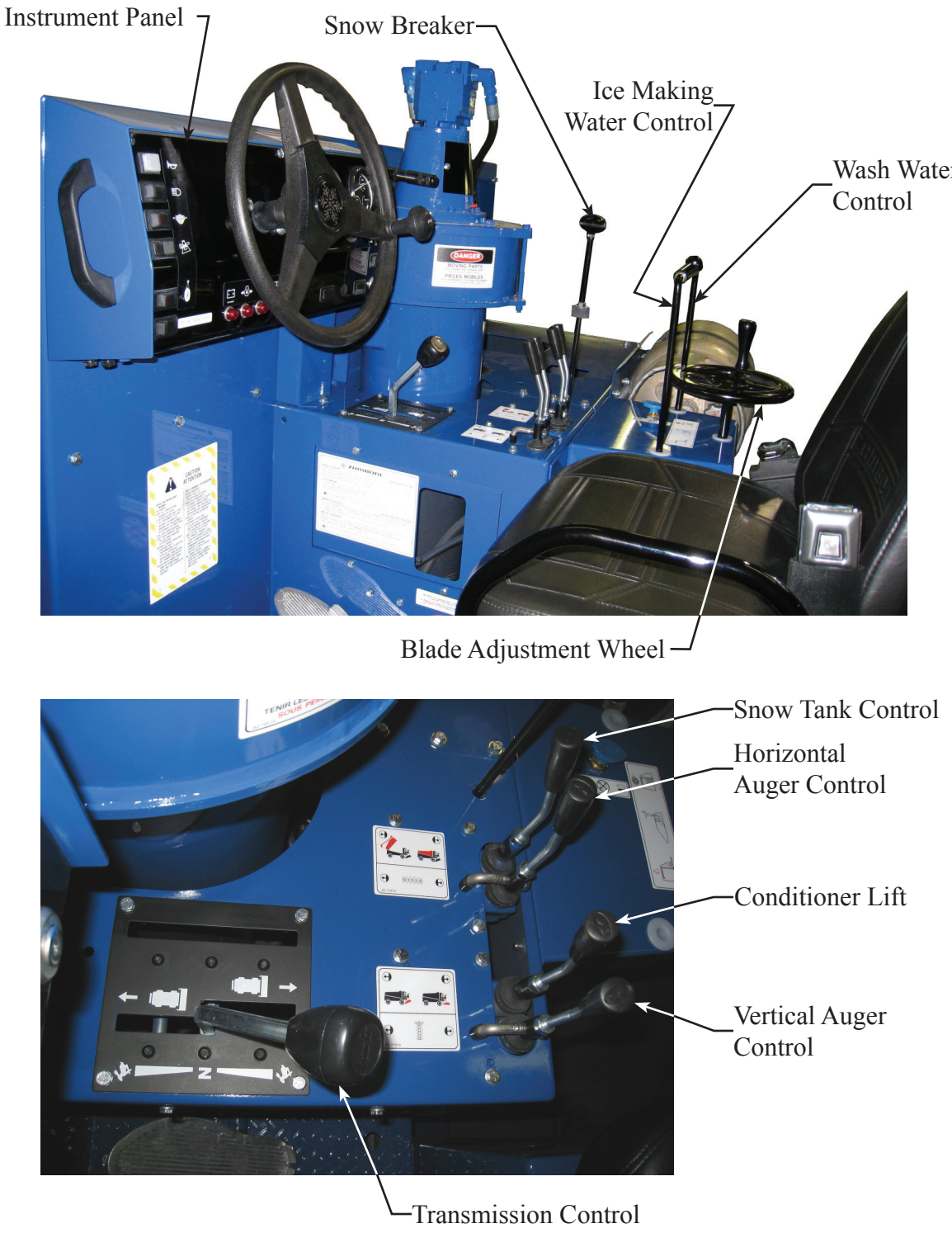


Figure 5.3 *Ice Resurfacer Components - Operator's Platform*

Power Package (Engine)

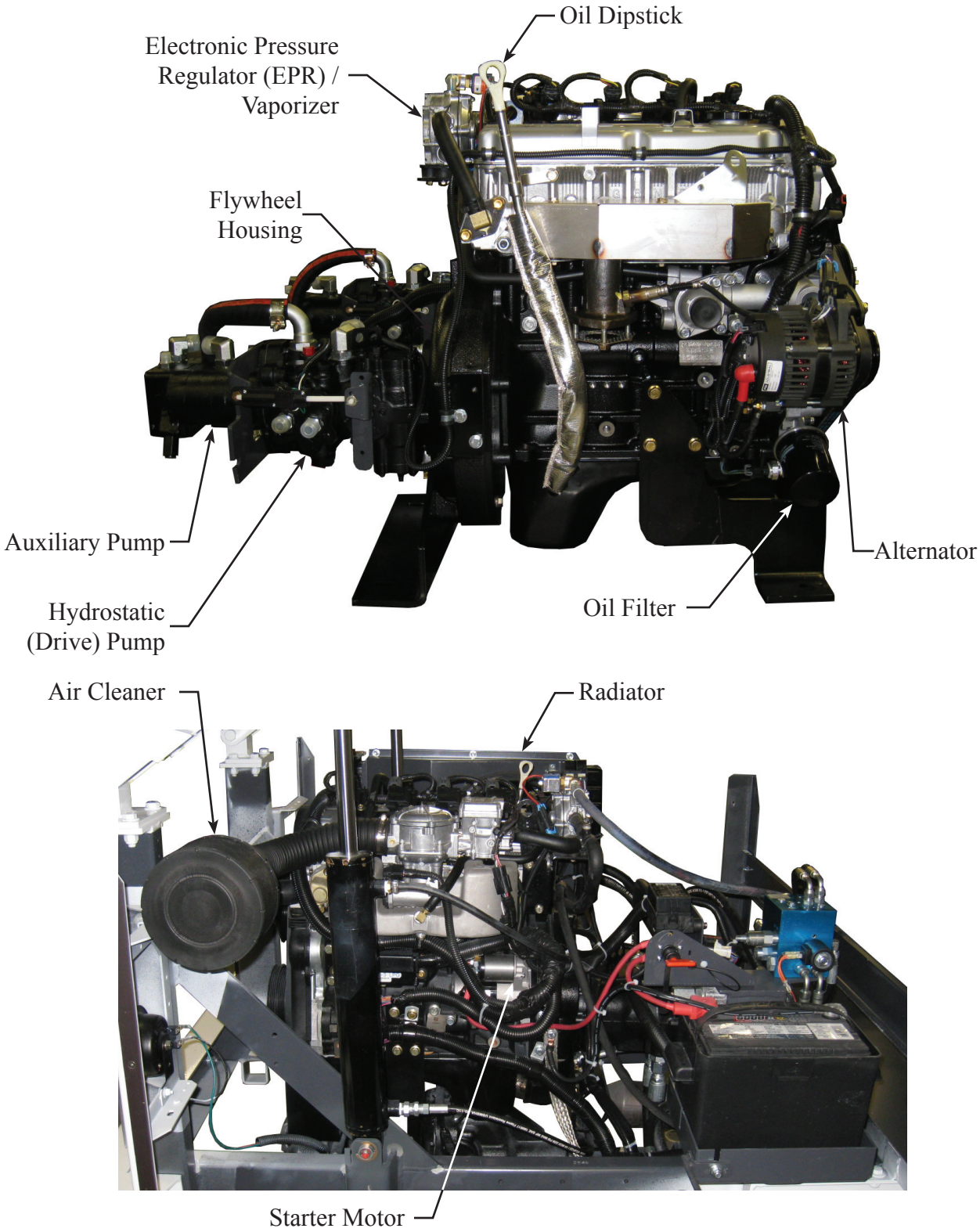


Figure 5.5 *Ice Resurfacer Components - Power Package*



Figure 5.6 *Ice Resurfacer Components - Snow Tank Safety Stand*

6.

Operator's Station

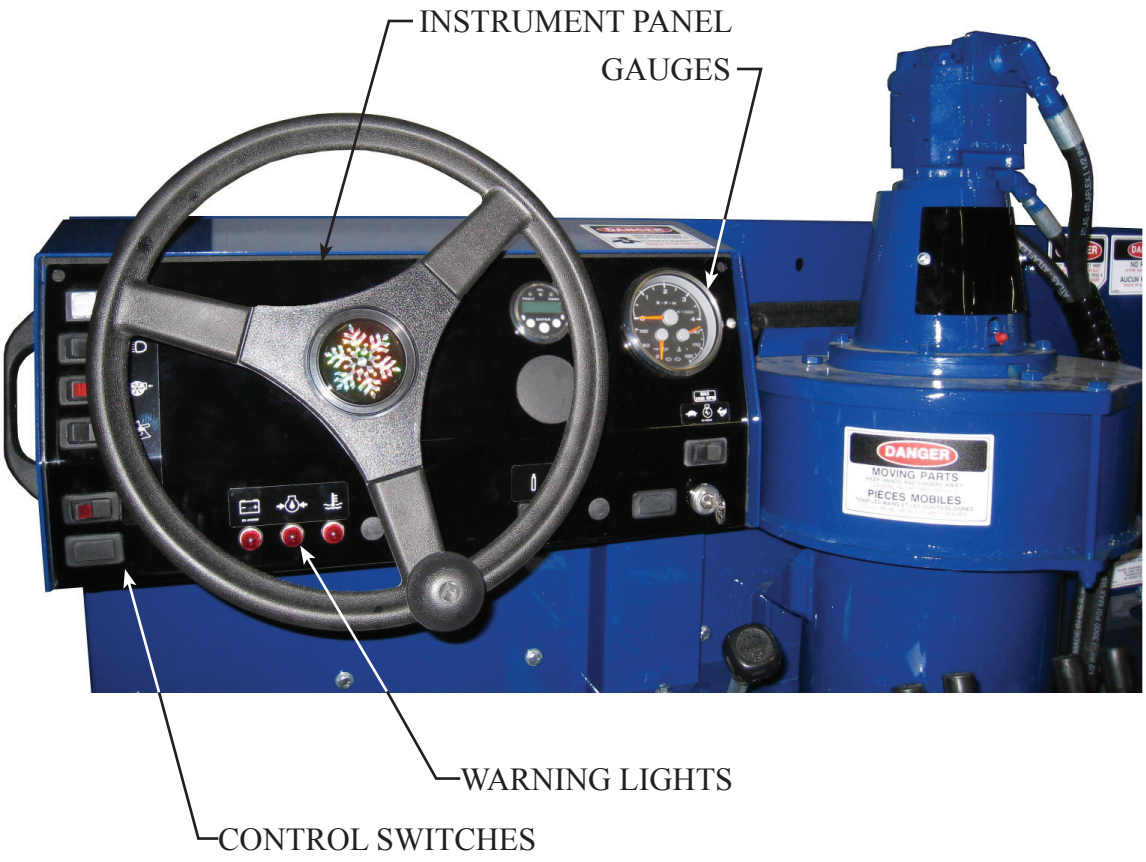


Figure 6.1 Operator Area

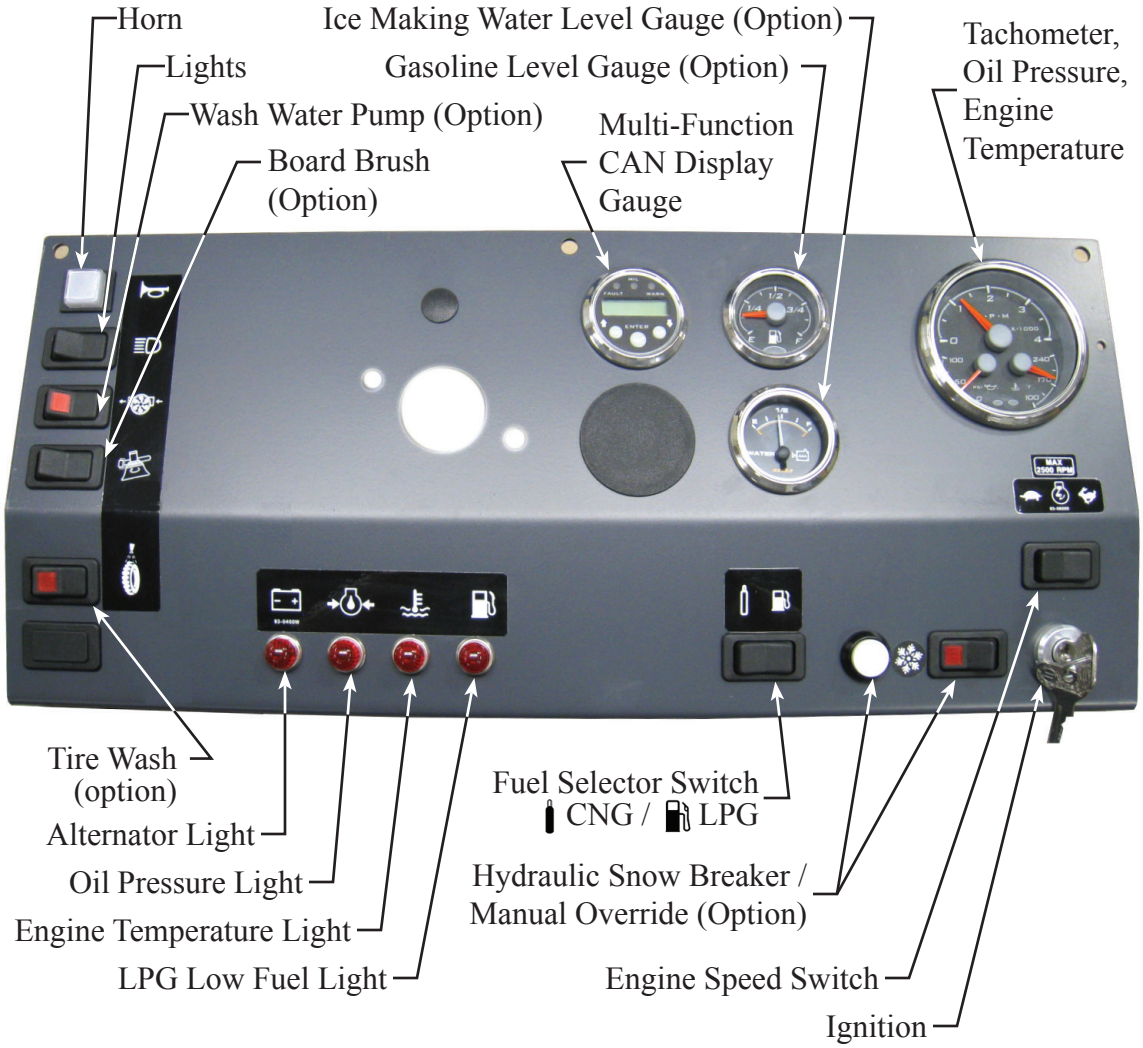


Figure 6.2 Instrument Panel

Tachometer, Oil Pressure, Engine Temperature Gauge

The tachometer, oil pressure, engine temperature gauge is used to view engine operating information in an easy to read analog format.

Multi Function CAN Display Gauge

This gauge gives the operator a variety of engine operating conditions which can be viewed by scrolling through using the tactile buttons on the gauge. The default display is engine hours.

Three lights (Warning, Fault and MIL) and an audible alarm are also built in. For further information on the CAN display gauge see page 6 - 6.

Operator's Station

Warning Lights, Buzzers, "Limp Home" and Automatic Shut-down Modes

The machine is equipped with a variety of warning lamps and buzzers that alert the operator of improper operating conditions. All warnings are indicated via a light, while some have an optional buzzer in addition. Extreme conditions that may cause severe damage to the machine also trigger power derate (50% throttle) or automatic shut-down modes. If "limp home" mode is activated, the engine's RPM will automatically be reduced and there is approximately 60 seconds of operation remaining before the engine shuts down. Use this time to turn off the augers, lift the conditioner and drive off the ice surface. If the ECU detects a low oil pressure condition, it will shut off the engine immediately.

Function	Light	Buzzer	Lamp Test On Start	Limp Home /Shut Down	Remedy/Notes
Engine Oil Pressure	YES	YES	5 SEC.	Shut Down	Check oil level, call for service
Engine Overheat	YES	YES	5 SEC.	Limp Home & Shut Down	Turn off engine, check coolant
Alternator	YES	YES	YES	NO	Check battery voltage, alternator output
Malfunction Indicator Light (MIL)	YES	YES	YES	NO	See fault codes

In all cases, if you are unsure of the cause and remedy for a warning lamp or buzzer, please contact the Zamboni Company for more information.

Engine Oil Pressure

The low engine oil pressure functions in 2 stages. Stage 1 will be activated if the oil pressure drops below 6 psig when the engine is running at or below 1200 RPM or 8 psig at or above 1450 RPM. Stage 2 is activated if the oil pressure drops below 3 psig at any engine speed.

During stage 1 the ECM will take the following action:

- Turn on the malfunction indicator light.
- Turn on the low oil pressure light.
- Provide an audible alarm.

During stage 2 the ECM will take the following action:

- Turn on the malfunction indicator light.
- Turn on the low oil pressure light.
- Provide an audible alarm.
- Shut the engine down after 3 seconds.

The operator can cycle the key off and then restart the engine. If the pressure is still low it will again shut the engine off after 3 seconds.

If you have an engine warning or shutdown due to low engine oil pressure, you must take the following actions:

- Verify the correct oil level and oil grade in the engine.
- If the condition persists, contact a mechanic for engine service.

Engine Overheat

The engine ECM turns on the high coolant water temp light when the engine reaches a temperature of 220°F for 15 seconds or more. The light has been set artificially low to warn the operator of a possible overheat condition and give as much time as possible to get off the ice prior to a coolant overflow condition or an engine shutdown condition.

The light may also come on during engine startup following a recent resurfacing or ice maintenance, if the engine was shut down at full engine RPMs and not idled down for the required 30 to 60 seconds to allow the coolant to circulate thus cooling the engine. If this (heat soak) happens run the engine at idle and the light will go out immediately after the coolant temperature goes below 210°F. To prevent this from occurring in the future, ALL operators should be instructed to idle the engine for 30 to 60 seconds immediately prior to parking the machine as recommended on the operating instruction decal.

The engine has built within it a high coolant water temperature shutdown feature. It operates in 2 stages.

- At Stage 1 (220°F) it will set a code, turn on the malfunction indicator light, provide an audible alarm and limit throttle opening to 50%.
- At this point the high coolant temperature light will be on.
- At Stage 2 (239°F), it will set a code; turn on the malfunction indicator light and shut down the engine after 3 seconds.

The operator can cycle the key off and then restart the engine. If the temperature is still above 239°F it will again shut the engine off after 3 seconds. It is very important that the operator get the machine off of the ice after stage 1 activates.

Check the coolant level in the radiator when it is safe to do so. Check the radiator for signs of leaking or damage. Check coolant flow for proper operation.

Alternator

The machine is equipped with a low alternator output indicator light. The light will be activated whenever the battery charge voltage is low. Check that the alternator is working correctly and that the battery is fully charged. Replace either of these items as necessary.

Machine Operation Switches

The switches located on the dash operate the following machine functions:

- Horn
- Lights
- Board brush (if equipped)
- Wash water pump (if equipped) (illuminated)
- Tire wash system (illuminated) (if equipped)
- Advanced water system (illuminated) (if equipped)
- Snow tank light
- Automatic snow breaker (illuminated) (if equipped)
- Low fuel warning light (if equipped)
- Engine RPM
- Fuel selector switch (if equipped)

Each switch has a label indicating its function.

The machine's ignition key switch has the following switch positions: off, on/run, start, accessory.

Ice making water gauge

This optional gauge shows the water level in the ice making water tank.

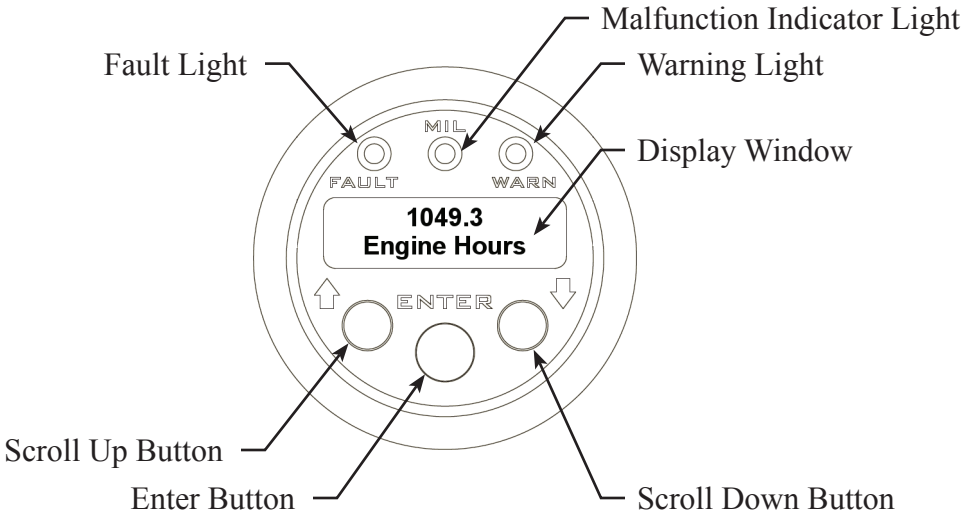
Multi Function CAN Display Gauge

The Multi Function CAN Display is connected to the engine control module (ECM) via a CANbus wiring harness. It displays real time values of the following engine parameters: Engine Hours (default display), Oil Pressure, Engine RPM, Coolant Temperature, Battery Voltage and Manifold Temperature. These parameters can be viewed by using the scroll up / down buttons on the gauge.

Further to the parameter display, the CAN display also contains the Malfunction Indicator Light (MIL), the Warning Light and the Fault Light. The MIL, the Fault Light and the Audible Alarm will be activated whenever the engine control module (ECM) determines that the engine is operating outside of its desired parameters. The operator must note this condition in the machine log and have the machine serviced immediately. Continued operation of the machine with the MIL illuminated may cause severe engine damage and may lead to dangerous operating conditions.

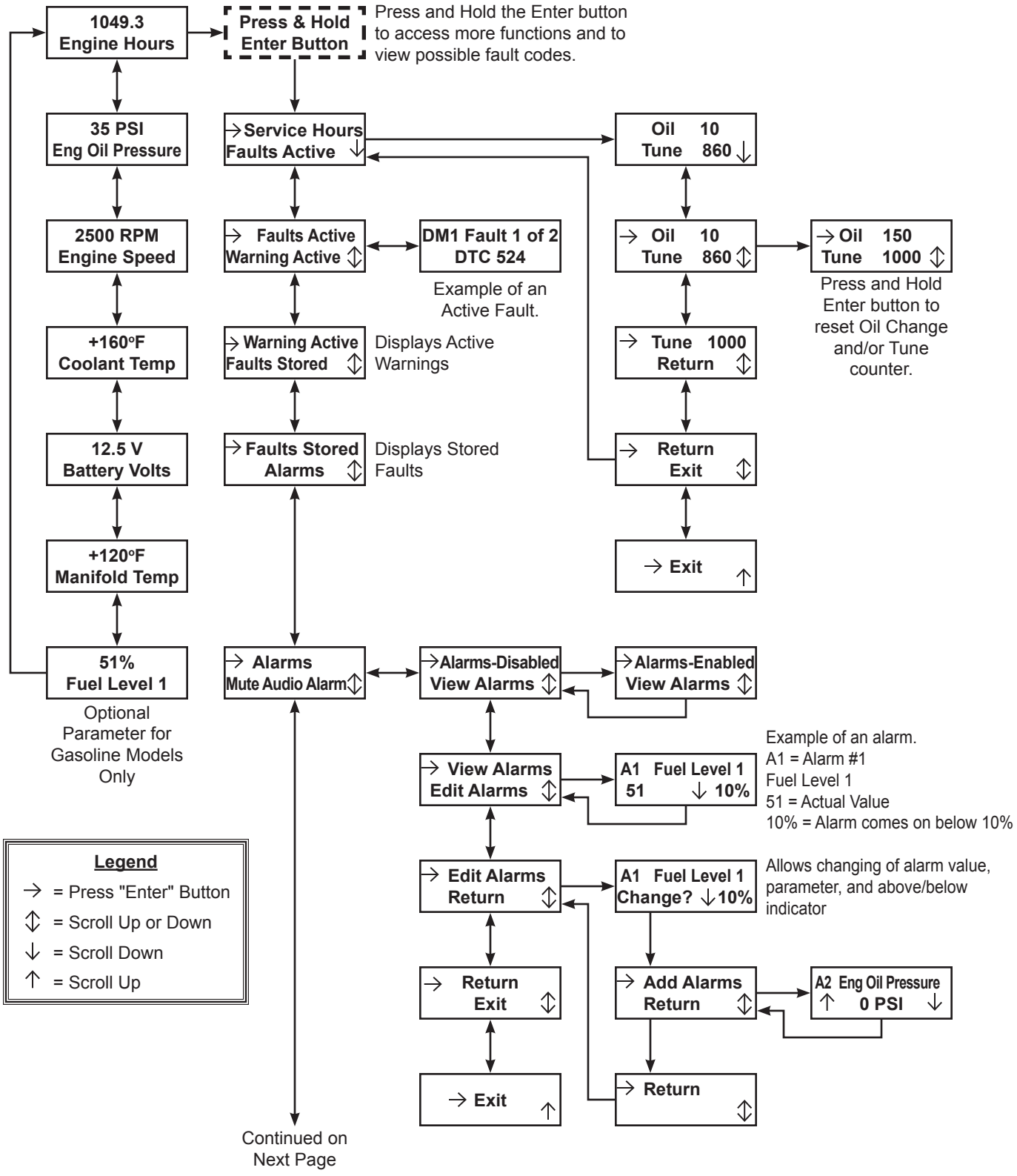
Along with the MIL, a Diagnostic Trouble Code (DTC) will be shown in the display window of the gauge. DTC codes are used to help diagnose the cause of a fault. A list of DTC codes and their meaning is in the engine service manual.

Service intervals have also been programmed into the Multi Function Display. The hours will count down to zero from the preset interval. Upon reaching zero either "Need Oil Change" or "Need Tune-Up" will be shown in the display window. At serial number 526-10806 and above the Warning Light will blink to provide further notification to the operator. Press any button to acknowledge the service notice and turn the light / notice off. Service interval reminders will be displayed at start-up until they are manually reset after completing the recommended service. The oil change interval is set for 150 hours. The tune up is set for 1000 hours.

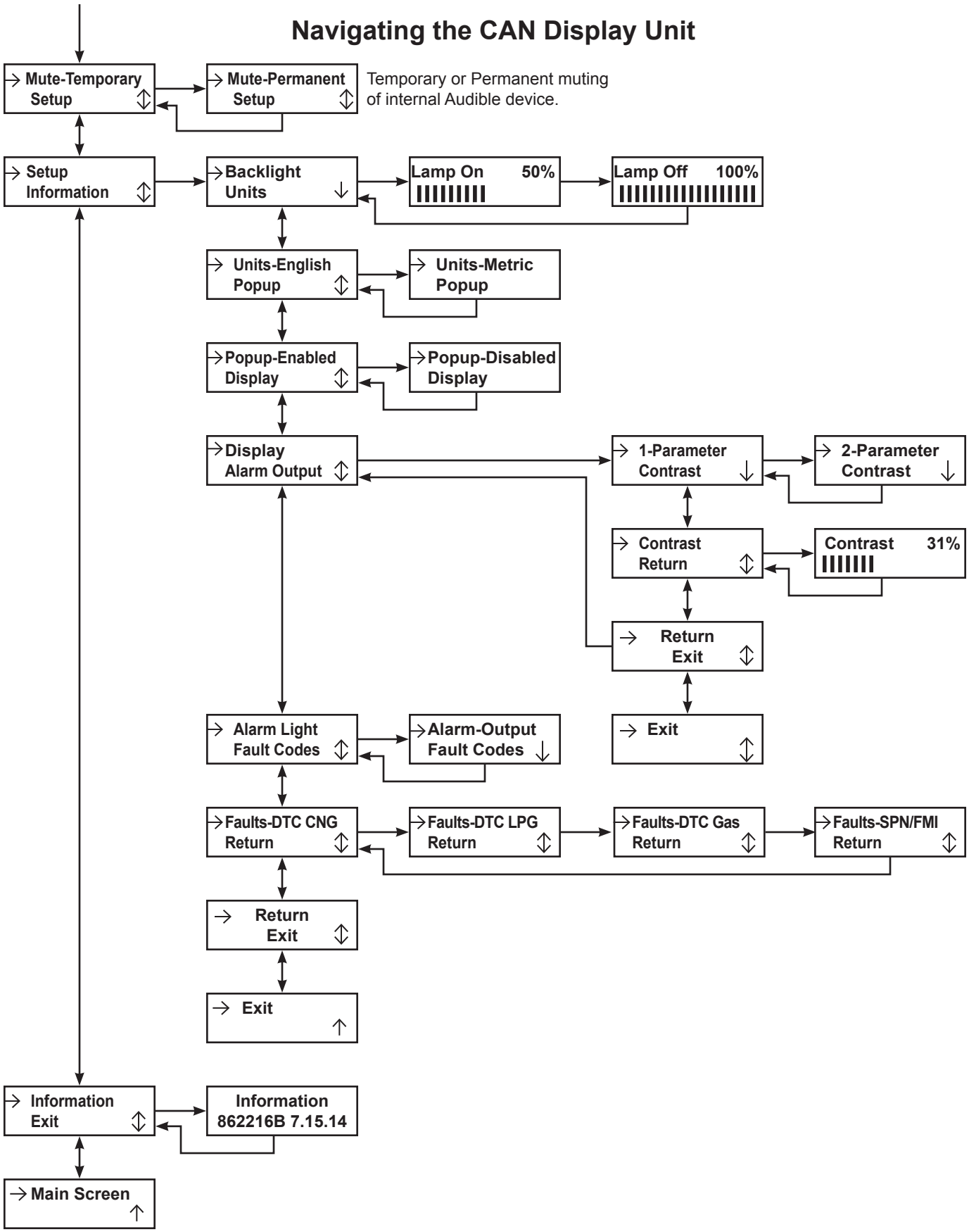


The flowchart on the following pages gives step by step direction on how to navigate the various functions of the gauge.

Navigating the CAN Display Unit



Navigating the CAN Display Unit



7.

Operating the Ice Resurfacer

Driving the Machine

Before Driving the Machine



Do not operate this machine before reading and understanding these instructions and also viewing the instructional video.

Always operate the machine with the seat adjusted to allow for proper foot pedal action.



Always operate the machine from a seated position. Do not stand up to operate!

NO RIDERS OR PASSENGERS ARE PERMITTED!

Do not operate the machine if the guards are not in place or broken.

Machine Controls

The Zamboni ice resurfacer has familiar automobile-style driving controls.

The ice resurfacer is steered with the steering wheel. It has two foot pedals, one that controls the direction and acceleration and the other one is a brake pedal. It has a hydrostatic transmission that allows for smooth operation.

The machine's ground speed is controlled through the use of the accelerator foot pedal and the engine RPM. The further the hydrostatic foot pedal is depressed, the faster the machine can travel. The engine speed (RPM) is controlled by the engine speed switch (See Figure 7.1).

Hydrostatic Transmission

The hydrostatic transmission is an infinitely variable hydraulic pump and motor combination. The machine's speed is selected with the acceleration foot pedal (Figure 7.1). The direction of travel is selected using the transmission control lever. The transmission must be in **neutral** for starting the engine.

The machine is equipped with a neutral start safety switch which prevents the engine from starting if the hydrostatic pump is not in neutral. The transmission lever must be in neutral and the foot pedal should be in the off position.

Engine Speed

This machine is equipped with an electronic governor. The engine speed is controlled via an engine speed switch (See Figure 6.2). The engine speed is increased by pushing the right side of the rocker switch. The engine speed is decreased by pushing the left side of the switch.

The electronic governor maintains the engine RPM independent of the load of the engine.

See your 526 Service Manual for more details on the electronic governor. When resurfacing, the engine speed should be set between 2200 and 2500 RPM. The operator must not "lug" the engine by operating with the engine RPM set too low. This will impair the operating and resurfacing performance and may damage the engine.

Machine Speed

The direction of travel for the ice resurfacer is determined by moving the transmission control lever as indicated in Figure 7.2. The engine speed should be maintained near 1500 RPM for travel and between 2200 and 2500 RPM for resurfacing.

The speed of the machine is controlled by the combination of engine RPM, the position of the transmission control lever and the accelerator foot pedal.

Accelerator Foot Pedal

The accelerator foot pedal activates the hydrostatic transmission. Depress the foot pedal and the machine will travel in the direction selected by the transmission lever. The foot pedal controls the amount of power applied from the transmission. Full pedal travel will produce full speed, maximum acceleration, etc. Lifting your foot from the foot pedal will slow the amount of power delivered from the transmission and will slow the machine. The hydrostatic transmission will act as a braking system when you lift your foot from the pedal. You can also apply the foot brake.

Throttle Switch and Hydrostatic Transmission Control

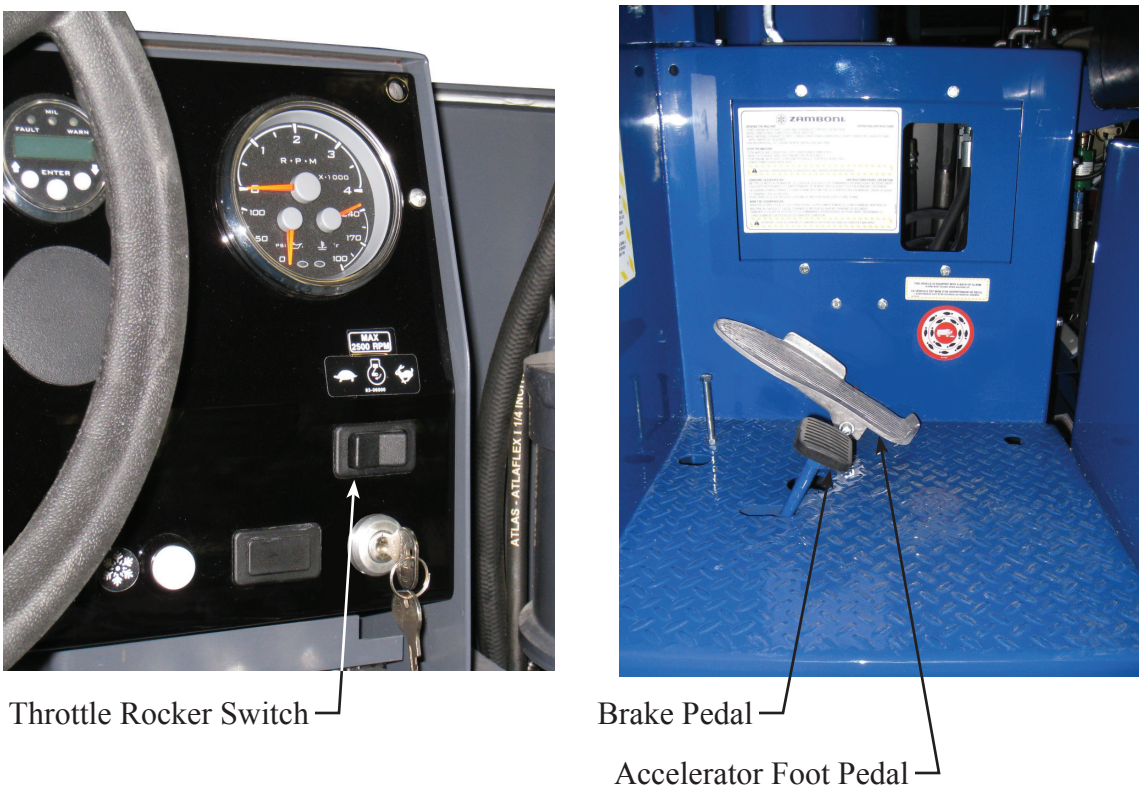


Figure 7.1 *Throttle Controls*

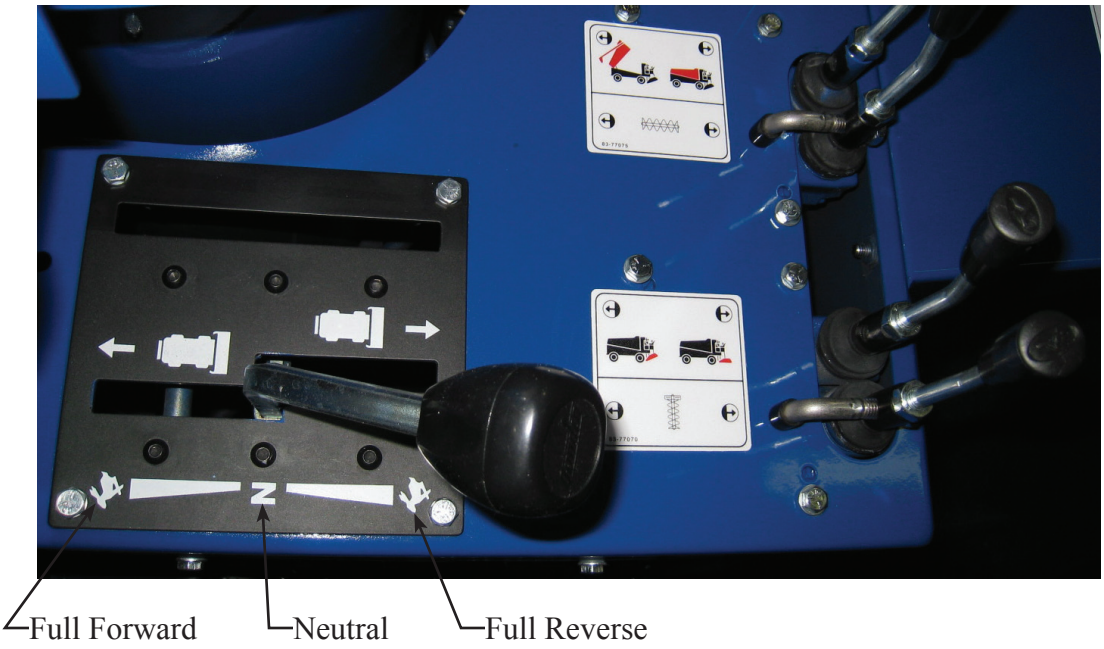


Figure 7.2 *Transmission Control*

Resurfacing Procedures

The Zamboni ice resurfacing machine produces the finest quality ice surface known. It eliminates the labor-intensive and time consuming methods of the past which were required to maintain an ice surface. You will produce higher quality ice if you read, understand and practice the following information.

Historically, the best sheet of ice produced prior to the development of the Zamboni ice resurfacer was usually produced by the following procedures:

- A tractor-pulled planer shaved the ice to remove as many skate grooves as possible.
- The accumulated shavings were plowed or scraped to a snow pit located next to the ice surface.
- Dirt and contaminants were flushed out of the remaining skate grooves by washing the entire ice surface using a heavy hose and nozzle.
- Large hand squeegees were used to push the dirty water to the snow pits.
- Using a hose, the final coat of water was sprayed or flooded onto the ice surface to complete the resurfacing operation.

The Zamboni ice resurfacer performs all of these tasks easier and more efficiently. The ice is shaved, shavings collected into the snow tank, the ice is washed using the wash water system and the final coat of water is applied to the ice surface.

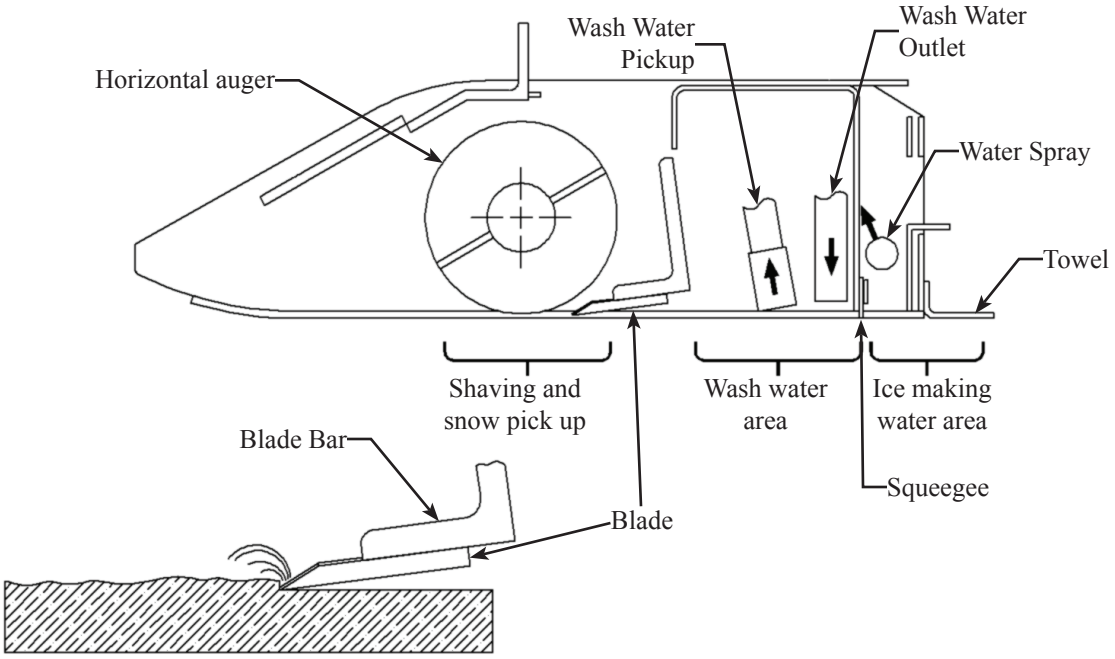


Figure 7.3 *Resurfacing Procedures*

Learning to Resurface

Before Going to the Ice

- Check the water level in the water tanks.
- The blade must be **level** (see Figure 9.5) and at the correct **angle** (see Figure 9.6).
- The conditioner must be raised completely.
- The towel bar should be attached if water is to be applied to the surface.
- NO RIDERS!

On the Ice

WARNING!



Do not operate the machine if anyone is on the ice.

The following operations may be performed while on the ice:

- Shaving or cutting.
- Washing the ice.
- Distributing ice-making water.
- Use of the board brush, edger, or other optional equipment.
- Removing excess water from the ice.
- Removing slush (mixed ice and water) from the ice.

Shaving or Cutting

- Proceed from the storage area onto the ice (see Figure 7.4).
- When starting the resurfacing pattern, it is recommended to start down the center of the ice surface so the functions can be started and blade adjustments can be made at some distance from the walls or boards.
- Before lowering the conditioner, make sure that the blade is "out-of the ice" or elevated so that it will not cut immediately upon lowering the conditioner to the ice surface. See the "Blade Control" information in Chapter 9.

  Note: To avoid catching the towel underneath the conditioner, lower the conditioner **only** while driving slowly forward.
- While on the ice surface and driving slowly forward, lower the conditioner and start the conveyors immediately.

To lower the conditioner:

Push down on the conditioner valve handle completely. Allow time for the cylinder to travel to the end of its allowable movement. You will normally hear a hissing sound produced by the hydraulic fluid passing over the relief valve when the conditioner has traveled fully down or up. When you hear this return the valve to neutral. This will insure proper down pressure has been applied. You should also feel the machine lift transferring weight to the conditioner.

To start the conveyors (augers):

Pull up on the conveyor handle. The operator should be able to hear the conveyors rotating. The operator should be able to observe the stream of snow through the viewing window as the snow tank fills with snow. The conveyors should be stopped before the snow tank is completely full to avoid plugging the vertical conveyor.

DANGER!



Turn off and remove the ignition key before adjusting, repairing, cleaning, or servicing the conveyors. If any part of the conveyor system becomes plugged or clogged, or appears plugged or clogged, the conveyor must be flushed out with water. Do not wear gloves or jewelry when performing these functions.

Flushing the augers out with water must be done only after the machine has been removed from the ice, the auger control valve has been returned to its neutral (off) position, and the ignition has been shut off.

Under no circumstances should the operator attempt to clean or clear the conveyor system or parts around it while the machine is on the ice or the power is on. To do so, exposes the operator and others to serious potential injury or death!

(Optional) Reversible Vertical Auger Operating Instructions

The vertical auger is equipped to rotate in both directions. The normal rotation for resurfacing is FORWARD, as shown by the information decal located near the valve operating handle.

Note: Since the vertical and horizontal augers are hydraulically connected in series, both augers will rotate in the reverse direction if the optional reverse auger funtion is enabled.

Reversing the auger rotation **may** clear a puck, water bottle, piece of hockey stick or other large objects that gets lodged in the auger flighting.

Reversing the auger rotation **will not clear a snow jam** in the auger, as the bottom of the auger is designated to pull in snow and not throw it out. The snow will just be compacted further.

To run the auger in the reverse direction, the valve stop needs to be relocated on the valve lever which will allow the valve to be actuated in the opposite direction.

The following steps should be observed when you decide to run the auger in the reverse direction:

- Loosen and rotate the valve lever stop out of the way.
- Being sure that the auger is not moving, actuate the lever to reverse the auger (auger lever moves down).
- To change auger rotation to forward, move the valve lever to NEUTRAL position, IF THE AUGER IS NOT RUNNING, pause a moment (2-3 seconds) then move the lever backwards for normal auger rotation. If the auger is running, you **MUST** move the lever into NEUTRAL and wait until the auger has come to a full stop.

CAUTION!



Never slam the lever back and forth rapidly to try to unclog something. You will definitely over pressurize the auger motor and cause breakage. Besides the unnecessary down time associated with this action, the damaged motor will not be covered under warranty. Damage created in this manner is very obvious upon inspection of the motor.

- Augers should be washed out between floods to prevent ice build up and snow jamming of the augers.
- After using the reversing feature, locate the valve lever stop in its original position (blocking the downward position of the lever).
- Depending on how long the reversing feature was used, you should check the auger bearings and the motor coupling screws to ensure that they are still tight. Reversing the auger may have loosened them off.
- The vertical and horizontal augers should be washed out at the end of the shift and the auger bearings immediately greased to displace any water accumulated during the day. Greasing the auger bearings once a day with high quality marine or boat trailer grease will ensure a long service life.

Typical Resurfacing Operation

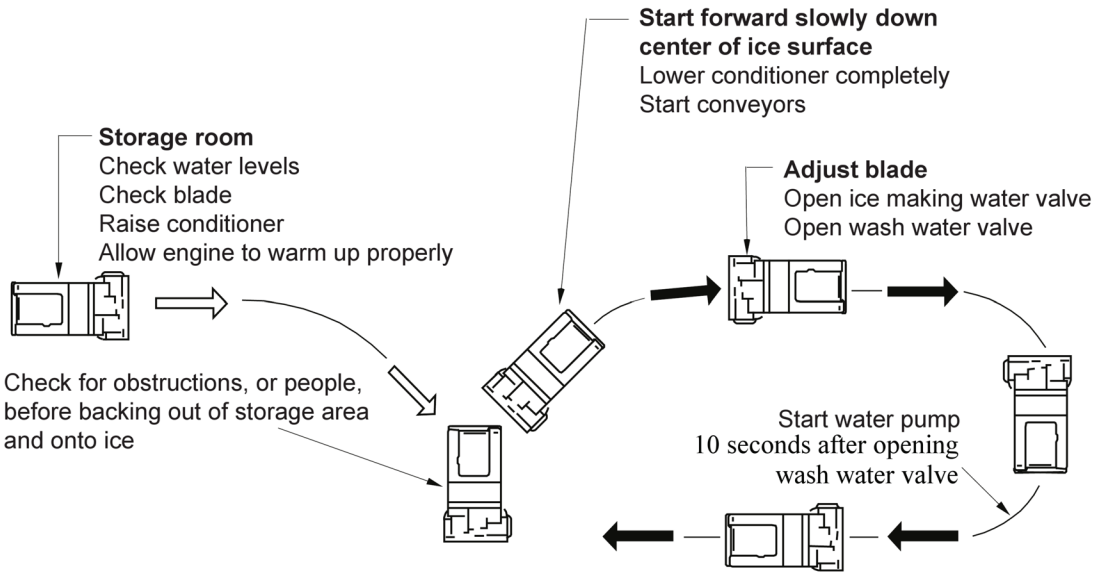


Figure 7.4 *Going onto the Ice*

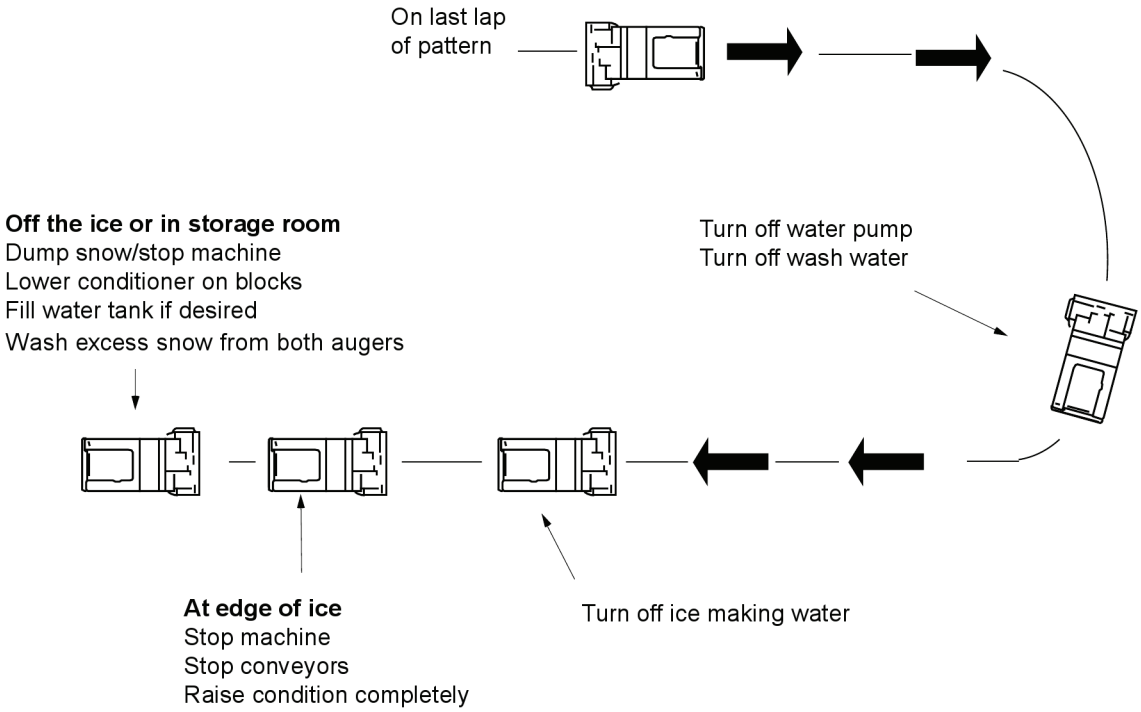


Figure 7.5 *Leaving the Ice*

Using the Snow Breaker

The operator should use the snow breaker several times a minute (or more as required). The snow breaker helps remove snow or ice that will build up in the corners of vertical auger opening. To use the snow breaker, push down on the snow breaker rod and release. Repeat this action several times.

The machine may also be equipped with an optional automatic snow breaker. It will perform the above action six times per minute once turned on via the rocker switch. It also has a manual override button should additional action be required.

Washing the Ice

The wash water system removes dirt and debris from the ice surface. Regular use of the wash water system will produce a clean, high quality ice surface.

Before using the wash water system, fill the wash water tank with cold water.

To use the wash water system:

- With the conditioner down, the horizontal conveyor rotating, and the machine moving forward, fully open the wash water valve. The wash water valve is located on the right side of the conditioner.
- Allow the valve to remain open for at least 10 seconds.
- Turn on the wash water pump switch. The red light in the rocker switch indicates that the wash water pump is operating. You should hear a click sound when the clutch on the pump engages.
- You will be able to see the wash water flow into the wash water tank screen basket located to the right of the operator's area. If there is no flow, turn off the pump immediately. Check the system for obstructions after the machine has been removed from the ice.
- When resurfacing has been completed, first turn off the wash water pump and then close the wash water valve. The conditioner can then be raised.

To avoid damaging the wash water pump impeller (caused by running it dry), always have water in the wash water tank and always wait at least 10 seconds after opening the valve before turning on the wash water pump. A burning rubber smell is an indicator of an impeller that has been or is running dry.

After each resurfacing, inspect and discard any debris collected by the wash water tank screen.

Periodically inspect the rubber squeegee located at the lower rear of the conditioner for tears and damage. Replace as required to maintain good wash water system performance. When replacing the squeegee be sure to trim the edge around the conditioner runner as tightly as possible. A sloppy cut will allow wash water to escape possibly creating a poor washing action and leaving streaks on the ice.

Ice Making Water

Using hot water for the ice making water will generally produce the best ice surface. Do not use water hotter than 140^0 F (60^0 C) as this could damage the plastic ice making water tank

To use the ice making water during a resurfacing:

- With the conditioner down and the machine moving forward, open the ice making water valve, located on the left side of the conditioner.
- You can regulate this water flow by opening and closing the valve. Some operators reduce the water flow in areas of the ice surface where they're driving pattern overlaps many times, such as in the corners, at the ends of the ice or the hockey goal areas to avoid leaving too much water.
- When the resurfacing is completed, close the valve before raising the conditioner.

Using the Board Brush

The optional board brush will help you produce a high quality ice surface by sweeping away the snow and ice that gathers at the base of the dasher boards, next to the kick plate, around the perimeter of the ice. It is important to sweep this snow and ice away before water is applied. This will prevent ice buildup on the boards. Ice buildup increases the need for ice edging.

The board brush is usually used during the first pass around the perimeter of the ice surface.

To operate the board brush:

- Activate the solenoid operated control valve by turning on the board brush switch on the instrument panel. (See Figure 6.2).
- At the activation of the control valve, the board brush arm will extend from the left side of the machine while lowering to the ice and the board brush motor will start.
- Carefully and slowly drive the machine parallel to the dasher boards so that the brush reaches the boards and sweeps the snow away from them. The roller on the board brush arm should contact the dasher boards.
- The machine should be positioned so the conditioner is as close to the boards as possible for two reasons. 1) Sweepings are directed into the auger area and collected by the conditioner. 2) The snow brush can then extend out on its own to the boards in the corners.
- When the roller is in contact with the dasher boards, the machine is in the correct location for best brush performance. The board brush arm has a spring action so that the machine's distance from the dasher boards can vary a little.

- After a complete lap of the ice surface next to the boards, retract the board brush by turning off the board brush switch. This will also stop the brush motor.
- Proceed with the other resurfacing operations over the remainder of the ice surface.

Removing Excess Water from the Ice

If there is an excess amount of water on your ice either from rainfall or excessive melting, it can be removed quickly by using the following procedure. The wash water tank should be emptied before this operation is begun.

*The ice resurfacer must be **MOVING** during these operations:*

- While on the ice surface, with the engine RPM at 2200 to 2500, lower the conditioner.
- Raise the blade off (or out) of the ice by turning the blade adjustment hand wheel counter-clockwise. This will allow the water to pass under the blade and accumulate in the wash water chamber. (Figure 7.3)
- Start the horizontal conveyor. This will provide power to the wash water pump.
- Start to drive, wait 5 seconds and then turn on the wash water pump. Remember that damage to the impeller will result if the pump is run dry for more than 30 seconds.
- Excess water from the ice should begin to flow into the screen of the wash water tank. After water from the wash system flows into the screen, begin to drive forward over the areas of excessive water.
- Fill the wash water tank.
- Turn off the wash water pump and conveyors.
- Drive off the ice and empty the wash water tank by opening the wash water valve on the conditioner. Lowering the conditioner may assist in getting additional water out.
- When through removing excess water from the ice, empty wash water tank and refill with fresh, cold water.

Diverting Excess Water from the Ice into the Snow Tank

Refer to page 8-4 for a full description of this additional feature.

Removing Slush (ice and water mixed) from the Ice

*The ice resurfacer must be **MOVING** during these operations:*

- While on the ice surface, with the engine at 2200 to 2500 RPM, lower the conditioner.
- Start the vertical and horizontal conveyors.
- Adjust the blade until it just touches the ice by rotating the blade adjustment hand wheel counter-clockwise.
- Drive forward over the areas of slush to be removed.
- Before leaving the ice surface and before raising the conditioner, continue running the conveyor augers for a few moments. This will help avoid freezing the slush in the tube between resurfacing. Stop the conveyors. Raise the conditioner and proceed to the dump area.
- After removing the slush from the ice surface and dumping it, flush the conveyors with water.

DANGER !



Turn off and remove the ignition key before adjusting, repairing, cleaning, or servicing the conveyors.

If any part of the conveyor system becomes plugged or clogged, or appears plugged or clogged, the conveyor must be flushed out with water. Do not wear gloves or jewelry when performing these functions.

Flushing the augers out with water must be done only after the machine has been removed from the ice, the auger control valves have been returned to their neutral positions, and the ignition has been shut off.

Under no circumstances should the operator attempt to clean or clear the conveyor system or parts around it while the machine is on the ice or the power is on. To do so, exposes the operator and others to serious potential injury or death!

Note: Whenever excess water or slush is being removed from the ice surface and conveyed into the snow tank, the Zamboni ice resurfacer ice making water tank **must be empty.** Water in both the ice making water tank and the snow tank may load the machine excessively.

Leaving the Ice Surface

When leaving the ice surface after finishing a resurfacing (or any other time): (See Figure 7.5)

- Close the wash water system valve. Turn off water pump (if used).
- → Turn off the auto snow breaker (if equipped).
- Raise the cutting blade out of the ice.
- → Turn off augers. ***This is especially important for machines equipped with the auto snow breaker option. Failing to stop both augers could result in damage to the snow chute and the actuating cylinder by contacting the horizontal auger.***
- Close the ice making water valve.
- With the augers stopped, raise the conditioner fully by operating the conditioner lift valve lever.
- If dumping outside, reposition the towel bar assembly to the raised position on the conditioner to avoid getting it dirty.

Off of the Ice

The following operations are performed off of the ice surface:

- Dumping the snow tank on a level surface if possible.

WARNING!



Raise the snow tank by operating the snow tank valve lever. After the snow has been dumped, lower the tank. If the tank is to be left up, the snow tank support stand must be properly placed to support the tank.

DANGER!



The snow tank should NEVER be raised while the conveyors are ON. When the snow tank is raised, the vertical conveyor blades are exposed and can cause personal injury if operating.

CAUTION!



The ice resurfacer should not be moved while the snow tank is being lifted or when the tank is in a raised position.

The machine should be level when raising or lowering the snow tank.

Operating Pointers

Resurfacing Patterns

The following diagram shows a good resurfacing pattern:

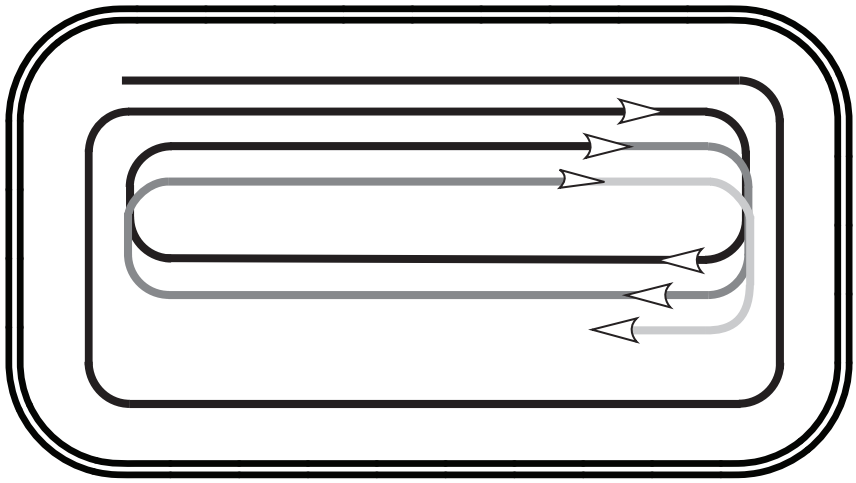


Figure 7.6 *Standard Resurfacing Pattern*

Sharp Turns

The ice resurfacer has a power steering system which contains a hydraulic relief valve. This valve only operates when the wheels are jammed up against an object or when the operator turns the steering wheel completely to the left or the right and holds it there against the stop.

If the steering is placed and held against the stop, the operator will notice a light, hissing noise coming from the instrument panel. This noise is made by steering relief valve opening and allowing hydraulic fluid to pass through it. The operator may also notice a drop in engine and conveyor speed. This is because extra engine power is being used to open the relief valve and push fluid through it.

Your turns will be much smoother and there will not be a drop in engine or conveyor speed, if you learn to drive the machine without placing the steering "on relief."

To do this you simply back off the steering stop a slight amount after you have turned the wheel completely and you hold the wheel steady in the turn. You can easily hear the relief valve open so you will know how tight you can hold a turn without always being on relief.

Excessive Loads

Taking too deep a cut with the shaving blade while driving the machine at it's full ground speed may place an excessive load on the engine. The engine RPM is maintained with only a small RPM drop while resurfacing normally. With an excessive machine load, the engine RPM may drop excessively which will cause the operation of the machine to suffer. If the engine RPM is dropping excessively, reduce the load of the machine by:

- Slowing the ground speed of the machine by lifting the foot pedal.
- Reducing the depth of cut.

8.

Water System

A. Pipe A (Figure 8.3) supplies water from the wash water tank to the conditioner.

B. Pipe B (Figure 8.3) carries water picked up in the conditioner to the suction side of the water pump. The pump then pumps the water to the top of the wash water tank, at which point the water can be recirculated through the filtering screen or pumped into the snow tank for quick disposal (Figure 8.5).

C. Hose C (not shown) attached to pipe C (Figure 8.3) supplies water through the ice making water valve from the ice making water tank to conditioner.

Wash Water Transfer (Option)

D. Valve D (Figure 8.4) may be opened to connect the wash water tank to the ice making water tank for additional ice making water capacity. Flush wash water tank clean before opening the valve when using this system. **Important:** Do not open this valve until the ice making water tank is half empty. Tank distortion can occur if the valve is opened while the tank is full.



Figure 8.1 *Water System*

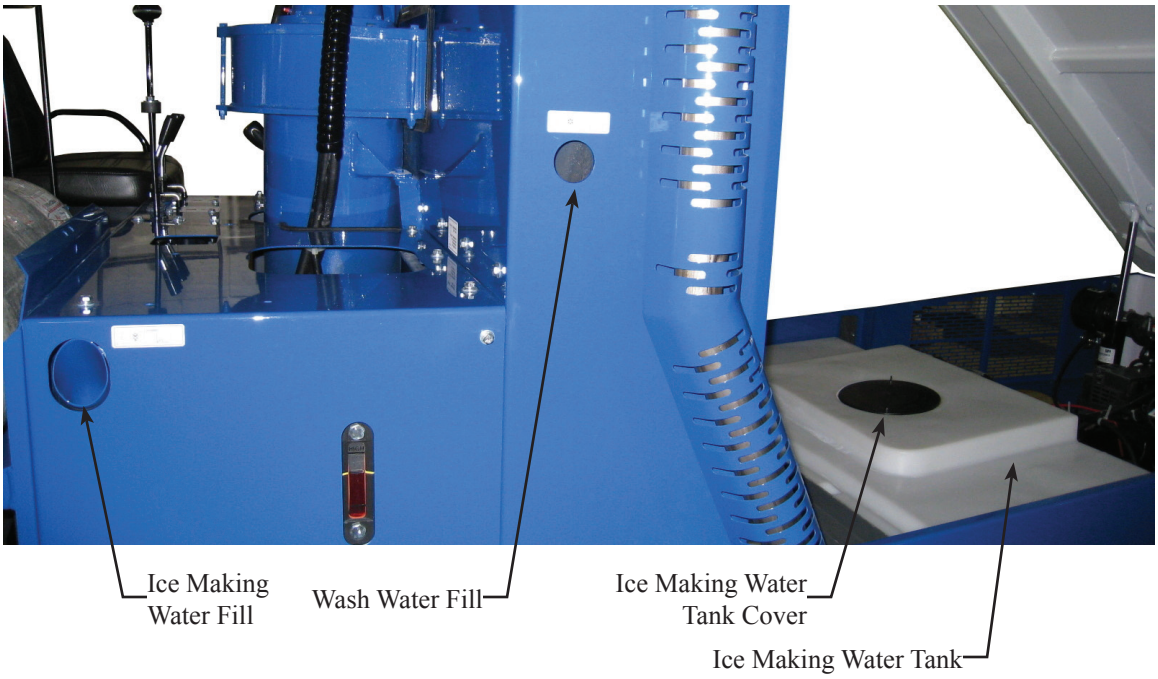


Figure 8.2 *Water Tank Inlets*

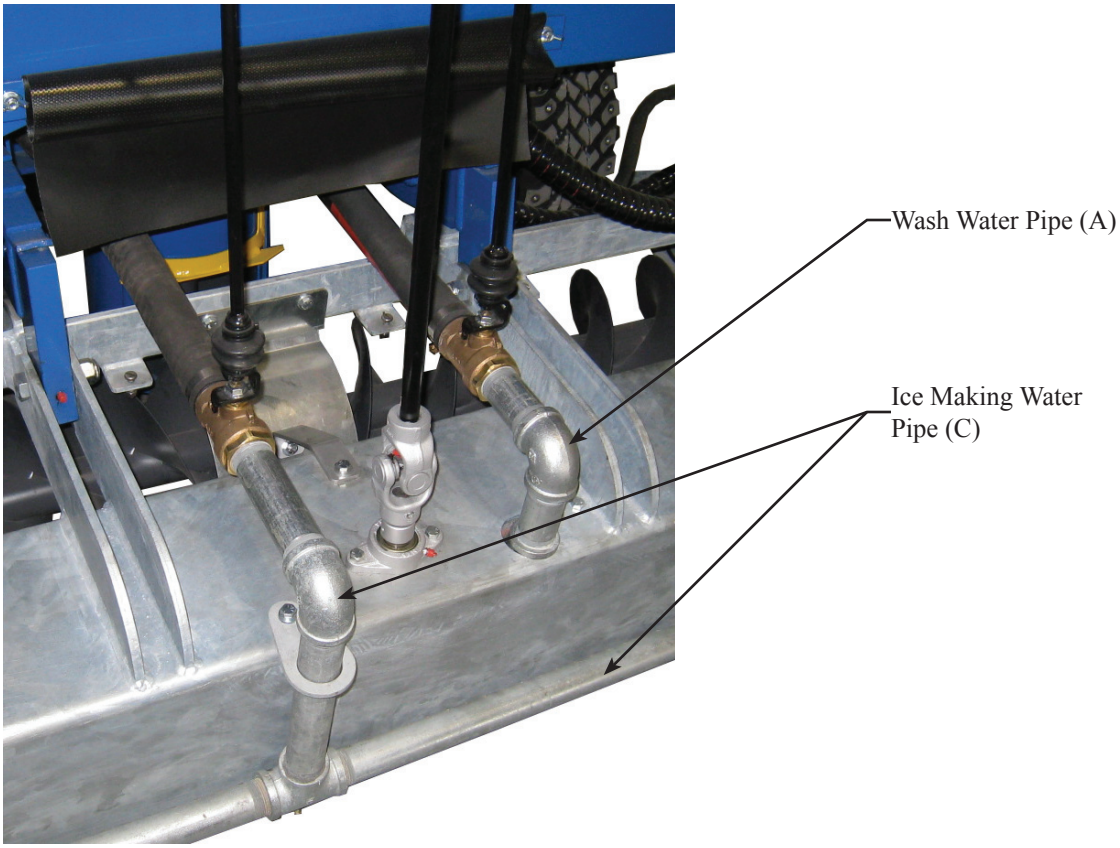


Figure 8.3a *Conditioner Piping*

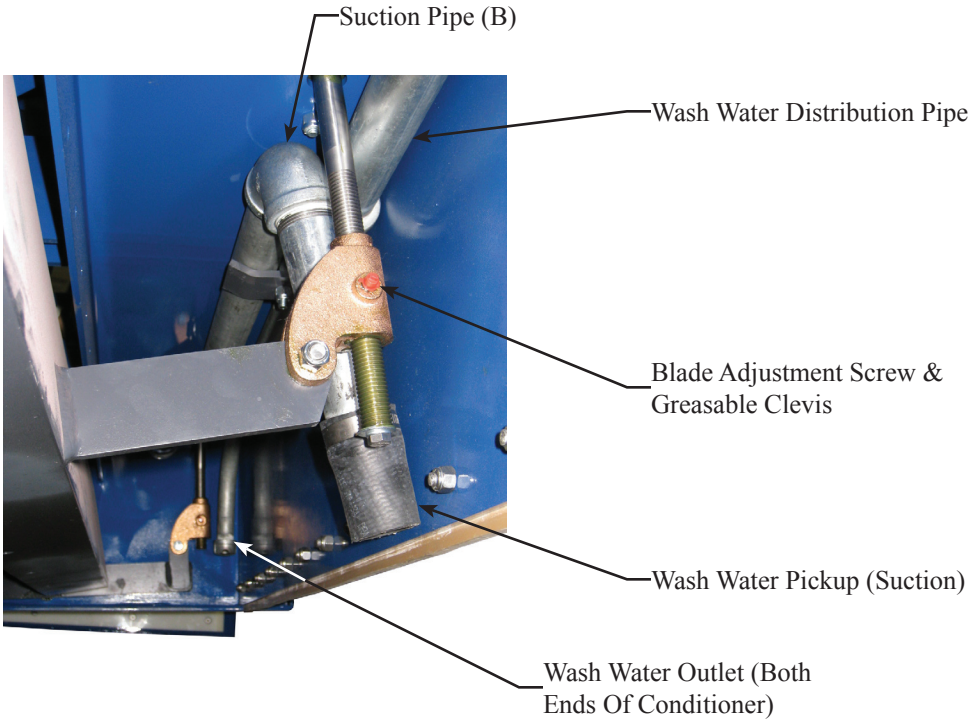


Figure 8.3b *Conditioner Piping*

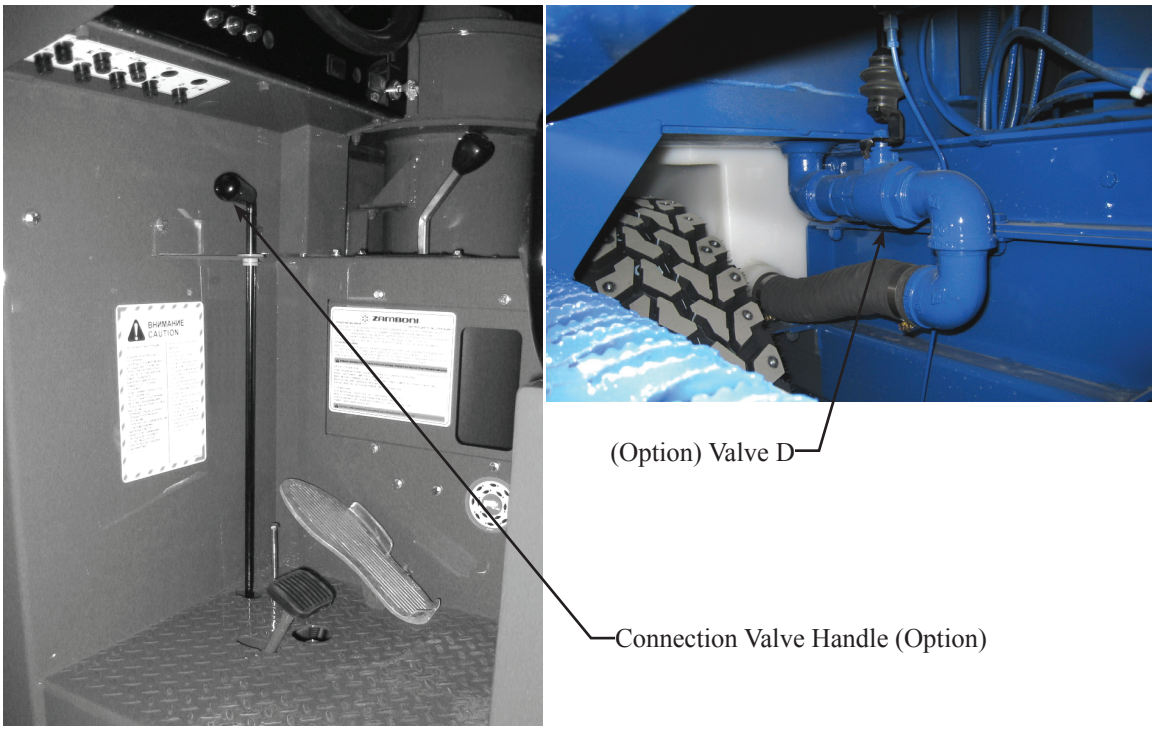


Figure 8.4 *Optional Connection Valve*

Diverting Excess Water from the Ice into the Snow Tank

If you do not wish to pump the excess water in to the Wash Water Tank, you may divert the water directly into the snow tank for easy disposal.

- Before starting to collect the excess water, rotate the Wash Water Return Pipe so it is directed into the Snow Tank instead of through the Filter Screen.
- Proceed to remove excess water as described in Chapter 7.
- When water starts to drain from the front of the Snow Tank, stop the Wash Water Pump and Horizontal Conveyor. Raise the Conditioner and move SLOWLY off of the ice to the appropriate dumping area.
- Raise the Snow Tank.
- Return the Wash Water Return Pipe to its original position once the excess water has been removed.

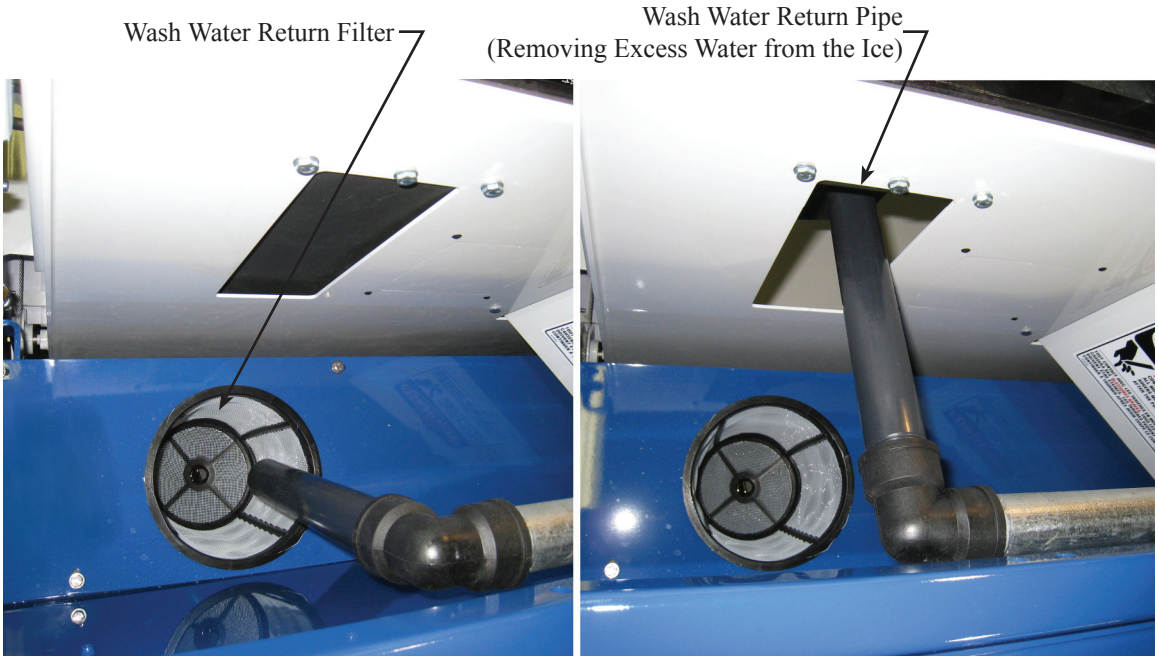


Figure 8.5 *Wash Water Filter and Bypass*

9.

Maintaining the Cutting Blade

It is essential that the blade be sharp if the resurfacer is to operate properly. Conditions vary from one rink to another, but experience has shown that normally the cutting blade is dull and needs re sharpening after 5 to 7 days of operations.

WARNING!



The cutting blade is extremely sharp and must be handled with care. The blade can INJURE whether sharp or dull. Use extreme caution! Use gloves and handle in scabbard as much as possible. Place blocks under conditioner while changing blade or reaching under conditioner.

Changing the Blade

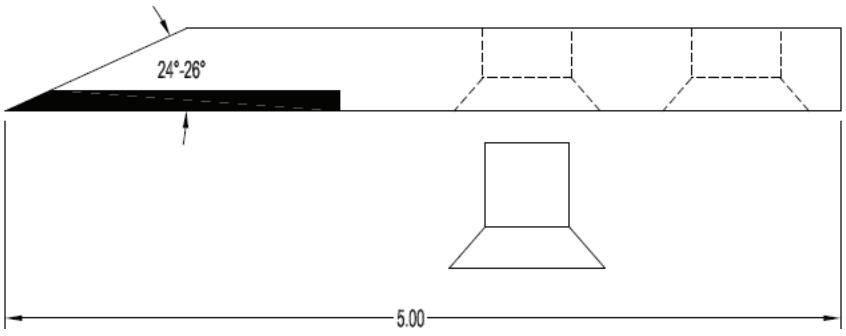
When changing your blade, see that the surface of the blade holding bar, to which the blade is bolted (Figure 9.4), is clean. Apply a small amount of lubricant, such as antiseize, to the bolts before assembly. This will help prevent galling, especially with stainless steel hardware. As the blade bolts are replaced, tighten the center nuts first and work toward the ends of the blade. The outside bolts are to be tightened last.

After removal, the dull blade should be sent immediately to be sharpened so that a sharp blade will be on hand at all times. For sharpening blades, contact a company that sharpens paper cutting knives for newspaper publishing firms.

Blades should be sharpened at an angle of 24^0-26^0.

New Blade

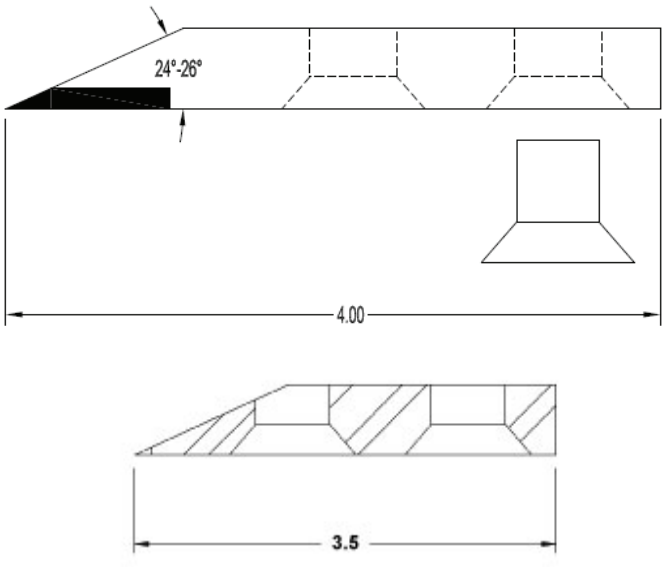
Use FRONT row of holes when blade is new.



Used Blade

Use REAR mounting holes when blade is 4" wide or less.

Note: Hardened material is exhausted when blade is approximately 3.5" wide. Discard blade.



Blade Controls on Conditioner

Blade Height Adjusting Screws level each side of the blade in relation to the runners.

Side Controls

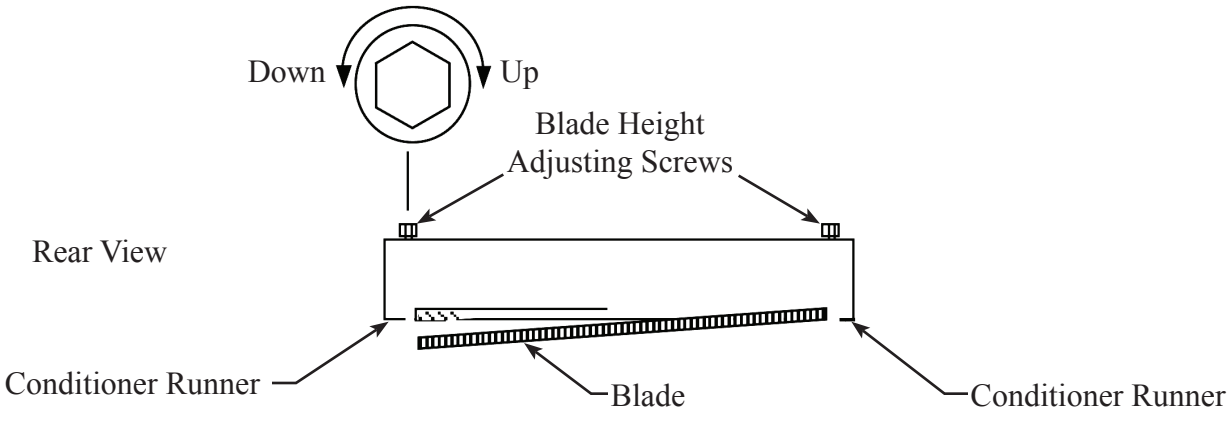


Figure 9.1 *Blade Controls on Conditioner*

Hand Wheel

The center hand wheel control varies the angle of the blade. If the blade is level with each runner, the hand wheel will make uniform changes in depth of cut.

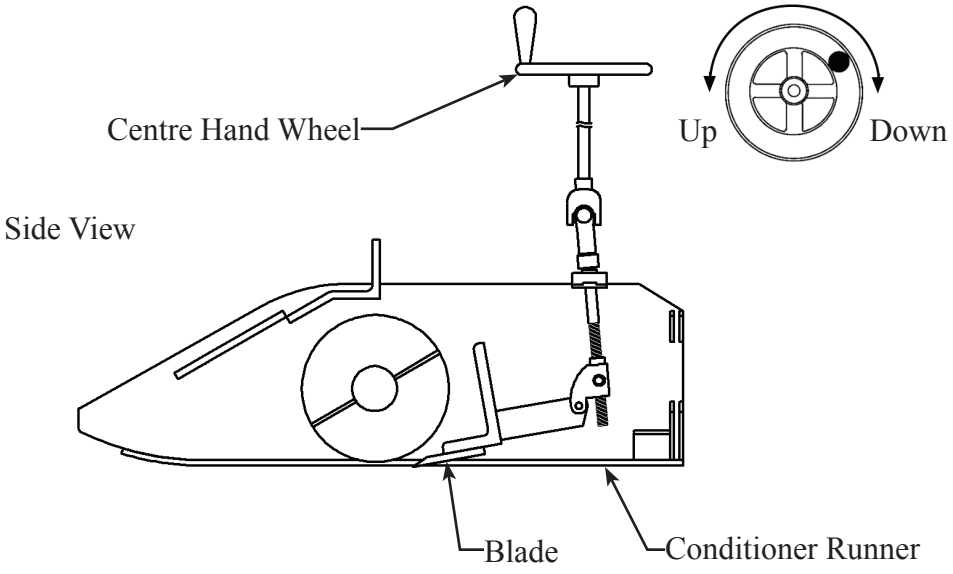


Figure 9.2 *Center Hand Wheel Control*

ARROWS SHOWN BELOW INDICATE ROTATION TO LOWER BLADE

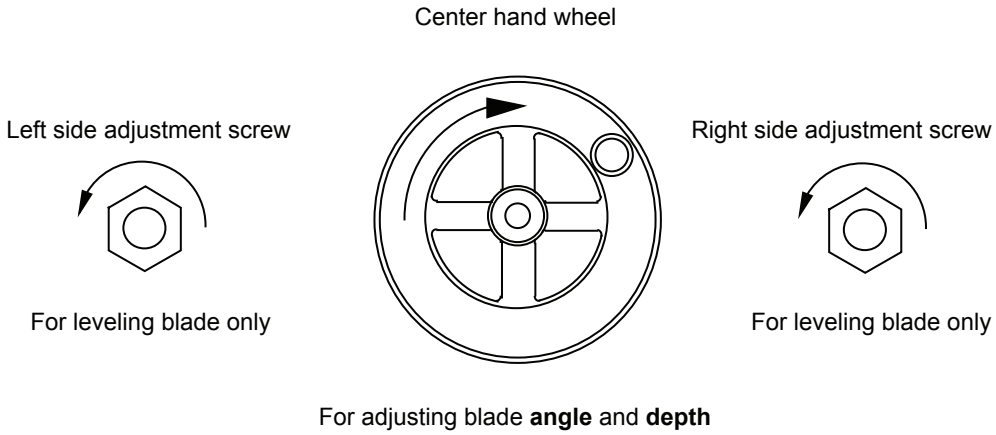


Figure 9.3 *Blade Adjusting System*

Holding and Securing the Blade

Blade Holding Bar (5K-78890) Blade (5K-33630)

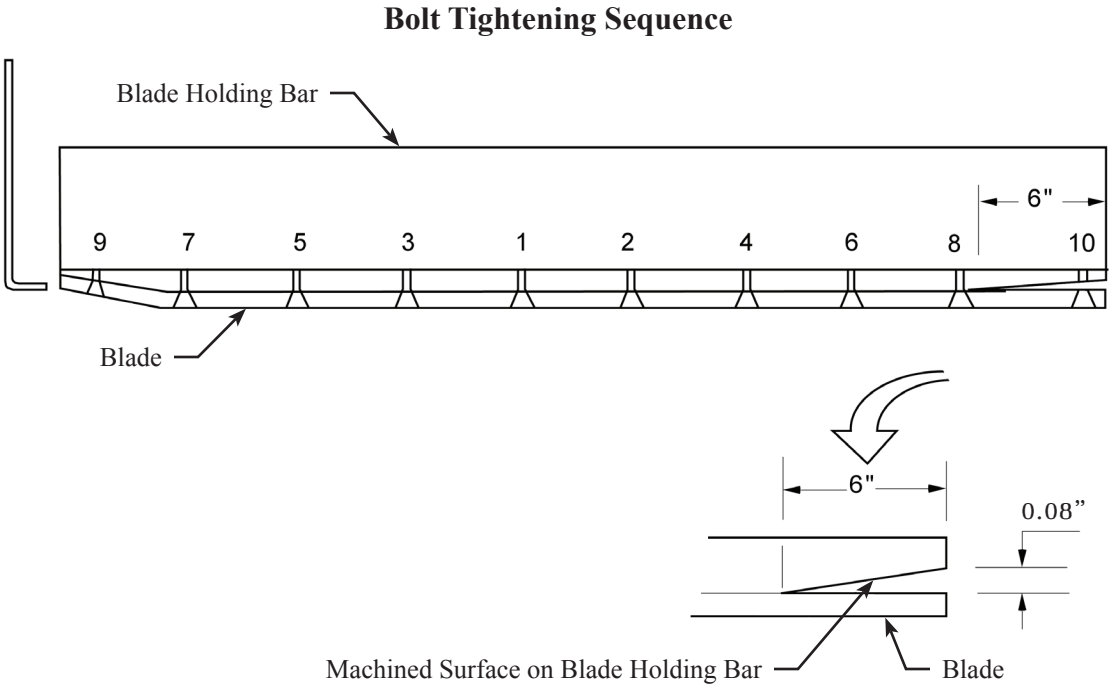


Figure 9.4 *Blade Holding Bar*

The ends of blade holding bar are machined upward approximately 0.08" so that one cut of the ice is feathered into the adjoining cut.

When attaching blade, tighten two middle bolts first, then work to outside.

WARNING!



Blade is extremely sharp. Avoid being cut by handling as much as possible in scabbard. Place blocks under conditioner while changing the blade.

Procedure for Setting Blade Angle and Depth of Cut

Step 1: Set blade at approximately 10^0 angle as shown on the blade angle adjusting plate (Figure 9.6) by rotating center hand wheel control.

Step 2: Align leading edge of blade EVEN with BOTH conditioner runners. It is quite helpful to use a flat washer, a large coin or the blade adjusting tool (Part No. 84-01100) to check this alignment as shown below.

WARNING!



Use caution as blade is sharp.

➤ **Note:** If blade is even with both runners, the majority of the blade will be about 0.08" below the runner as shown in Figure 9.4. Since a 0.08" cut would normally be quite a heavy cut, turn the hand wheel counter-clockwise to raise the blade slightly. (With a new 5" wide blade, 4 turns of the hand wheel is equal to approximately 0.08").

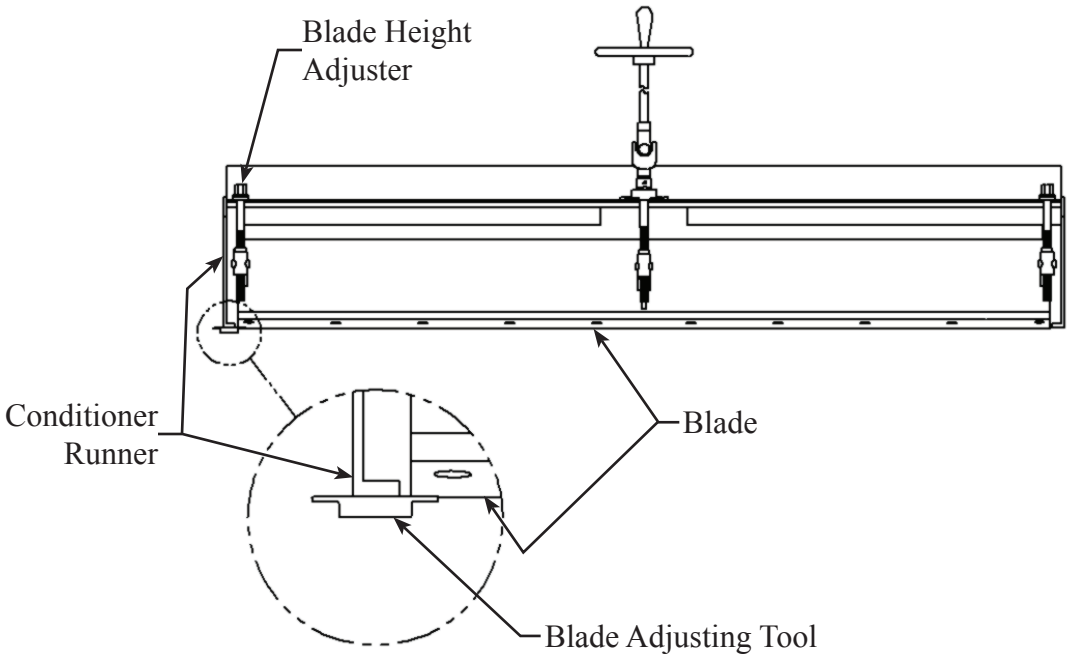


Figure 9.5 *Aligning the Blade with Conditioner Runners*

Blade Angle Adjustment Decal

A blade angle decal is located on the inside of the conditioner left side plate (see Figure 9.6). When the blade holding bar is lined up, parallel to the series of lines, the blade angle is approximately 10˚.

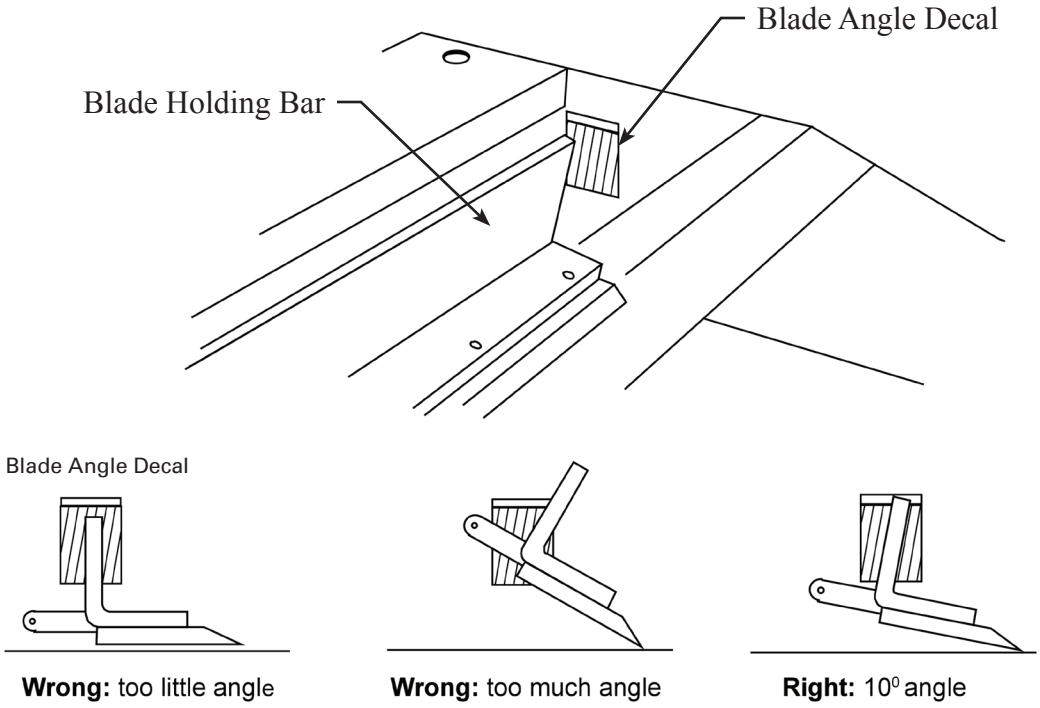


Figure 9.6 *Blade Angle Adjustment Decal*

The blade angle gauge (84-01100) may also be used. Place the angled face of the tool on the underside of the blade. When the flat edge of the tool is parallel to the runners, the blade is angled correctly (see Figure 9.7).

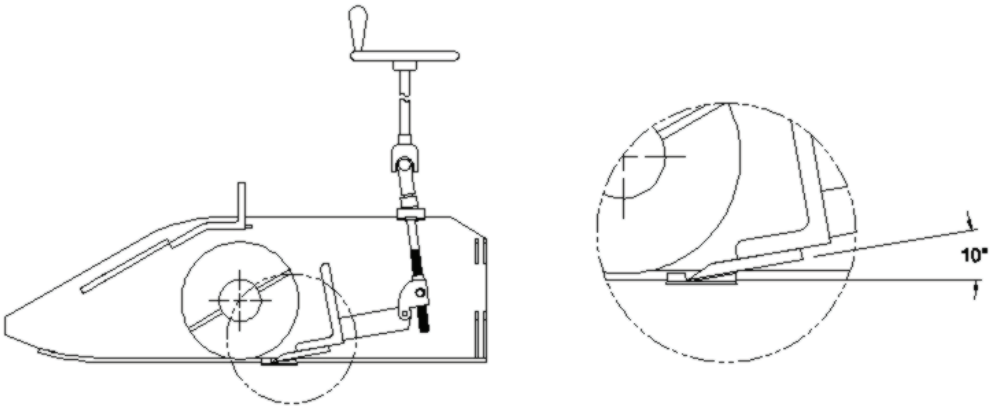


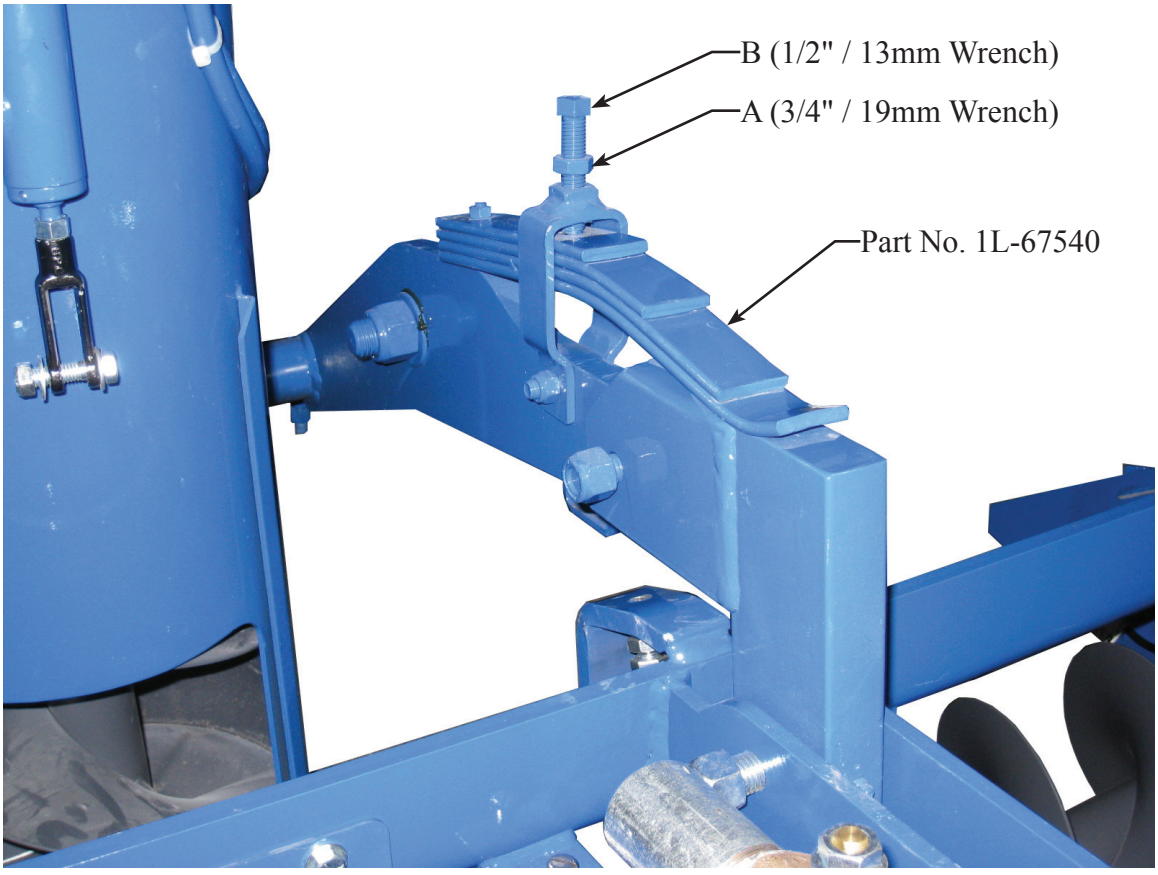
Figure 9.7 *Blade Angle Adjustment Tool*

Conditioner Down Pressure

The purpose of the conditioner down pressure system is to create downward pressure on the shaving blade thus increasing the shaving capability of the machine.

Down Pressure Spring Adjustment Procedure

- Lift Conditioner slightly off of the ground, just so it clears the floor.
- Loosen lock nut A.
- Loosen set screw B and tighten against spring "hand tight". Tighten with a wrench 3 to 4 complete revolutions.
- Tighten lock nut A.
- Repeat above procedure on opposite side.



Spring tension is now set for proper down pressure in operating mode. NOTE: Periodically check the condition of the down pressure springs (Part No. 1L-67540) to insure that they have not flattened out and taken a permanent set.

To extend spring life, the conditioner should be lowered between resurfacings onto floor blocks without applying down pressure (machine should not lift).

Using the Edger

During the normal operation of an ice rink, the edges of the ice surface have a tendency to build up.

If not removed, this buildup makes proper shaving of the rink perimeter difficult and may become a safety hazard to skaters.

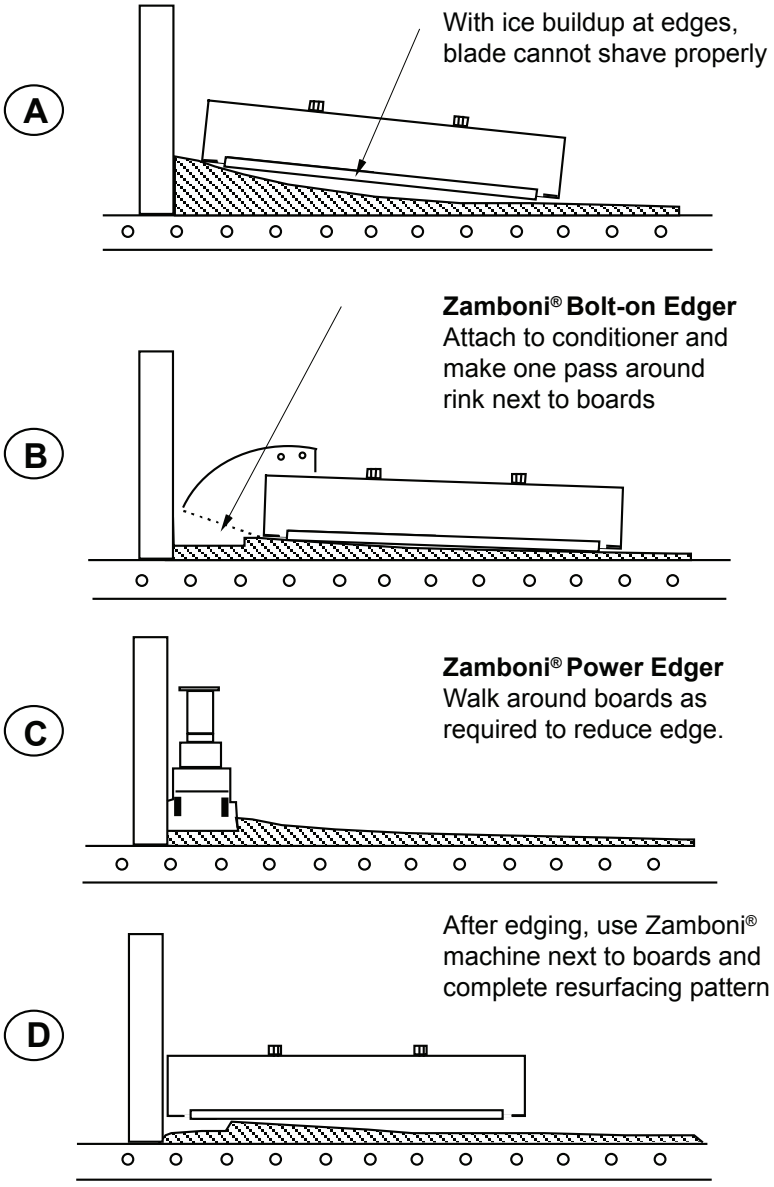


Figure 9.8 *Zamboni Edger*

Edger Attachment

Mount the edger on the driver's side of the conditioner and bolt tightly at the three locations shown below. The depth of cut on the edger blade may be adjusted by turning the two blade adjustment set screws.

Use the edger as a separate operation in the following manner:

- Start driving **slowly** next to boards. Fully lower conditioner, but raise the conditioner blade off of the ice, slightly, by turning the blade adjustment hand wheel counter-clockwise.
- Start conveyors.
- Take one or two laps along the side of your boards, keeping the edger against the railing or dasher.
- Remove the edger assembly.
- Resurface the ice rink completely in the normal manner.

CAUTION:

Drive slowly when using the edger. While using the edger, the front of the resurfacer will tend to pull into the wall.

Note: Keep edger blade sharp.

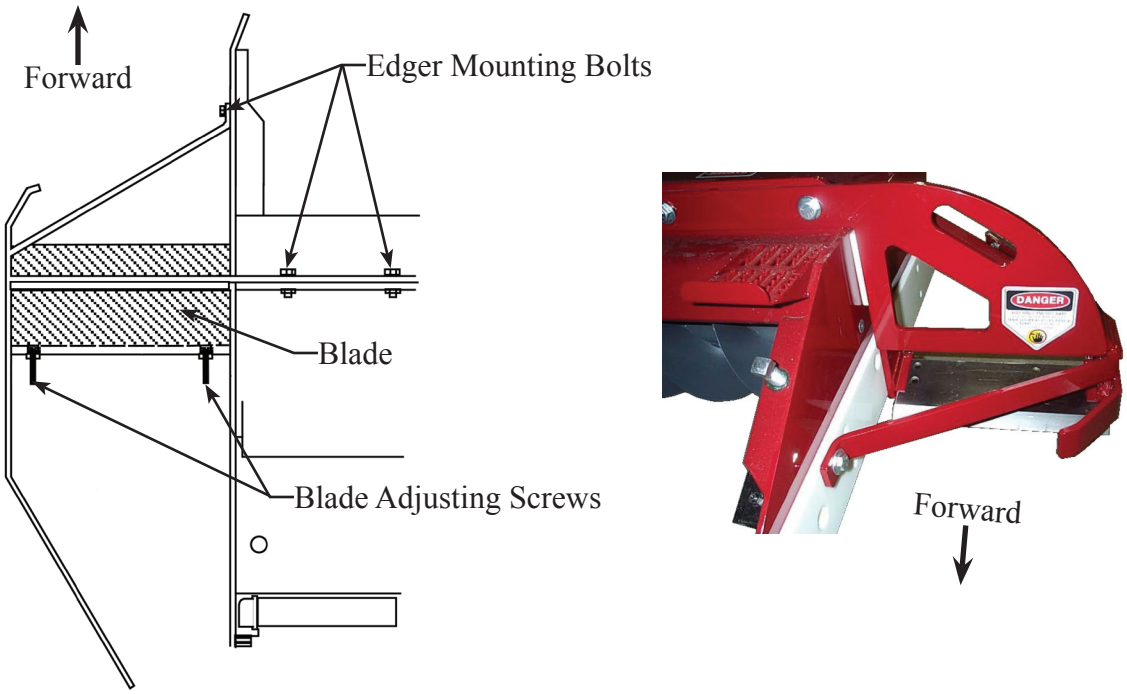


Figure 9.9 *Edger Attachment*

10.

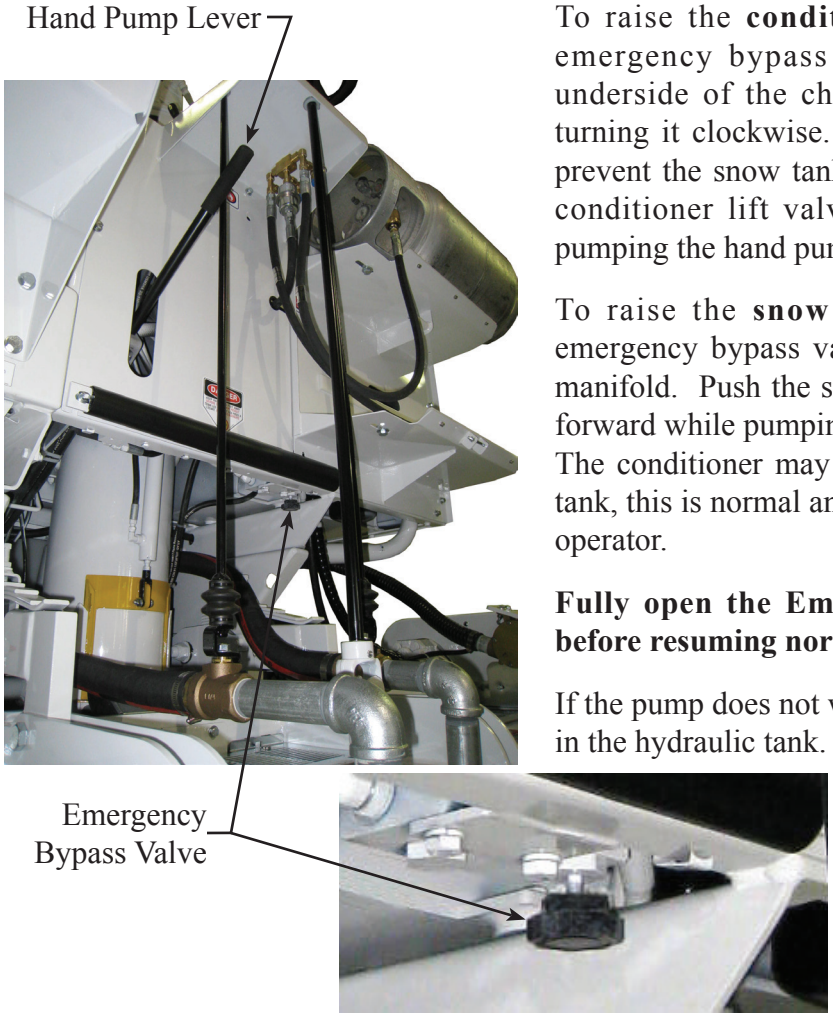
Emergency Operation

Emergency Hand Pump

The emergency hand pump is located in the driver area of the machine. The pump handle must be stored where the operator can find it in an emergency.

Purpose: The emergency hand pump is used to raise the conditioner or the snow tank in the event the engine will not operate.

Operating the Emergency Hand Pump



To raise the **conditioner only**; close the emergency bypass valve located on the underside of the check valve manifold by turning it clockwise. Closing the valve will prevent the snow tank from lifting. Push the conditioner lift valve lever forward while pumping the hand pump lever.

To raise the **snow tank**, fully open the emergency bypass valve on the check valve manifold. Push the snow tank lift valve lever forward while pumping the hand pump lever. The conditioner may lift as well as the snow tank, this is normal and should not concern the operator.

Fully open the Emergency Bypass Valve before resuming normal operation.

If the pump does not work, check the oil level in the hydraulic tank.

Figure 10.1 *Hand Pump Actuation and Emergency Bypass Valve*

Emergency Hydrostatic Bypass Valve

Should it be necessary to move the machine when the engine is not operable, the emergency bypass valve, located on the hydrostatic pump, must be opened (see Figure 10.3).

This valve is located on the hydrostatic pump and painted red for ease of identification, in the engine compartment. (See Figures 5.5 and 10.3 for location). To open the bypass valve, rotate it two turns counter clockwise using the adjustment tool (Part No. C4-00157 or 84-05870) supplied strapped to the suction hose.

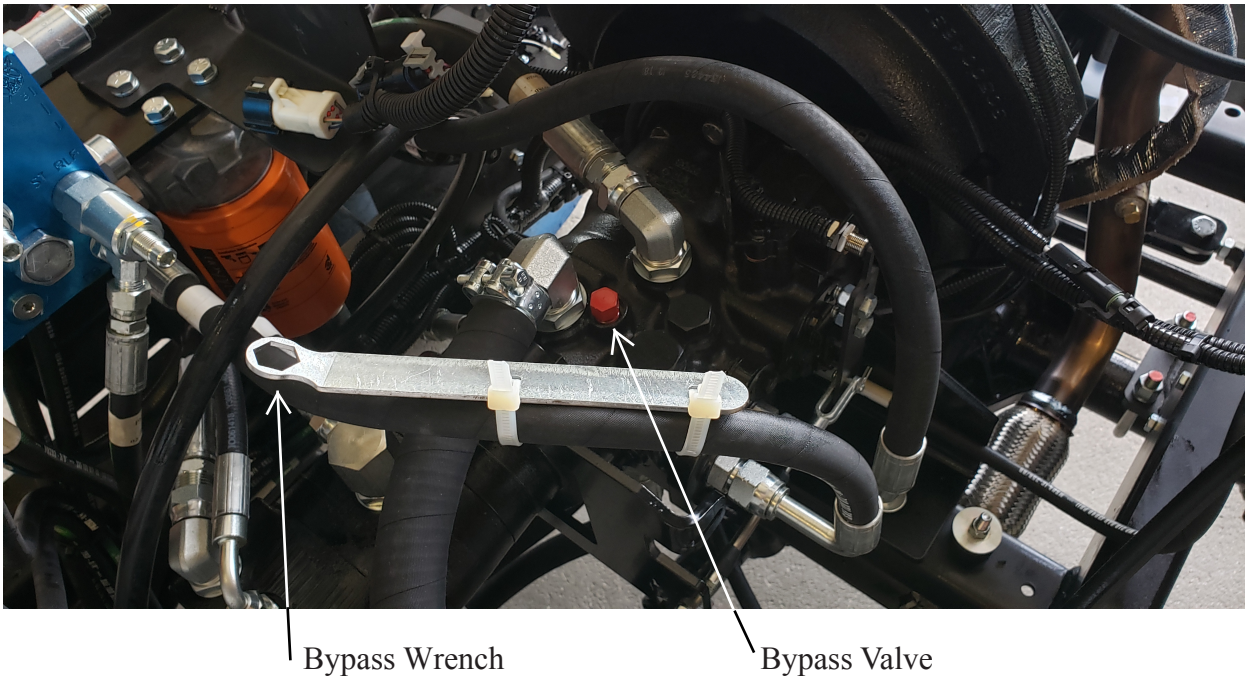


Figure 10.3 *Hydrostatic Pump Bypass Valve*

Notes:

- This system is to be used **ONLY** when pushing or pulling the machine **SLOWLY.**
- The bypass valve must be **FULLY** closed for normal operation.
- **Disconnect** the drive shafts for extended towing distances.
- **Fast towing must not be done.**
- Always close the valve once the machine is moved to the desired location. Re-attach the bypass tool to the hydraulic hose.

11.

Storage Information

Storage Room Information

Storage Area

Due to the great amount of water associated with this machine, it should be stored between resurfacing in a room heated above freezing. The room should be as close to the edge of the rink as possible and the surfaces between the storage area and the rink should be paved so dirt will not be tracked onto the ice. It is highly recommended that the storage room be on the same elevation as the surface of the ice to eliminate the need for a ramp.

Keep all sources of open flame, spark or other ignition such as stoves, furnaces, water heaters or any appliance using a pilot light away from fuel storage, fuel use and refueling areas. Under well ventilated conditions, a minimum distance of 20 feet from all ignition sources is recommended; however check your local bylaws as they will take precedence.

Floor Drain

The floor of the storage room should be pitched toward a floor drain of sufficient size to carry excess water and melting snow away from the machine.

Daily Storage (Indoor)

It is suggested that the snow tank be left in the lifted position for storage with the safety stand in place. This will allow for better air circulation for drying and will cut down on corrosion and rusting.

Summer Storage Tips

- Empty the water tanks. The ice making tank is made of polyethylene and will not present a corrosion problem. The wash water tank is made of steel with a zinc rich primer topped with a coat of epoxy. Remove the cover annually and insure the integrity of the paint has not been compromised. Touch up as required.
- As recommended in maintenance literature, completely lubricate the chassis and ice resurfacer parts to discharge any possible moisture from the bearing surfaces.
- If possible, store the machine in a dry enclosed area.
- If possible, park the machine on a flat and level surface. If the machine is parked on any kind of grade the wheels must be blocked with wheel chocks to prevent the machine from rolling away after it has been parked.
- If possible, raise the machine onto blocks thereby taking the weight off the tires.

ZAMBONI® CIRCLE CHECK LIST

The overall objective of the "circle check" is to evaluate the ice resurfacer and its systems, as well as the environment in which it will be operated.

Table of Contents:

- General Circle Check
- Circle Check Detail
- Front / Rear View Detail Diagram
- Side View Detail Diagram
- Top View Detail Diagram

General Circle Check:

- **Auger Systems** should be in good condition and free of snow, ice or obstructions
- **Fuel Supply** should be assessed prior to resurfacing
- **Towel** should be clean and free of tears
- **Squeegee** should not have tears or cracks and should have a snug fit from end to end
- **Ice Making Water** supply should be sufficient for resurfacing and have steady and even distribution from the pipe
- **Wash Water** should have steady flow and a clean basket/bag
- **Board Brush** should be firmly attached to the unit and show no signs of damage
- **Basic Fluid Levels** should be sufficient and not in need of change
- **Belts and Hoses** should be tight fitting without wear and with no signs of aging / leaking or any other need for replacement
- **Oil and Filters** should be monitored for replacement as recommended by the machine's service manual
- **Grease Horizontal Auger Bearings** on a daily basis
- **Drain Water Tanks** at the end of each day
- **Fuel Supply** should be turned off at the end of each day for fuel powered machines
- **Ventilation System** should be on and active and air quality tests should be performed per the facility's schedule
- **Check Under / Around / Behind** resurfacer and observe the condition of the path into and out of the ice resurfacer storage area
- **Snow Pit / Dumping Area** condition and path to/from the area
- **Evaluate Consumables** such as fluids and parts which may be show-stoppers if not on hand

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ZAMBONI® CIRCLE CHECK LIST

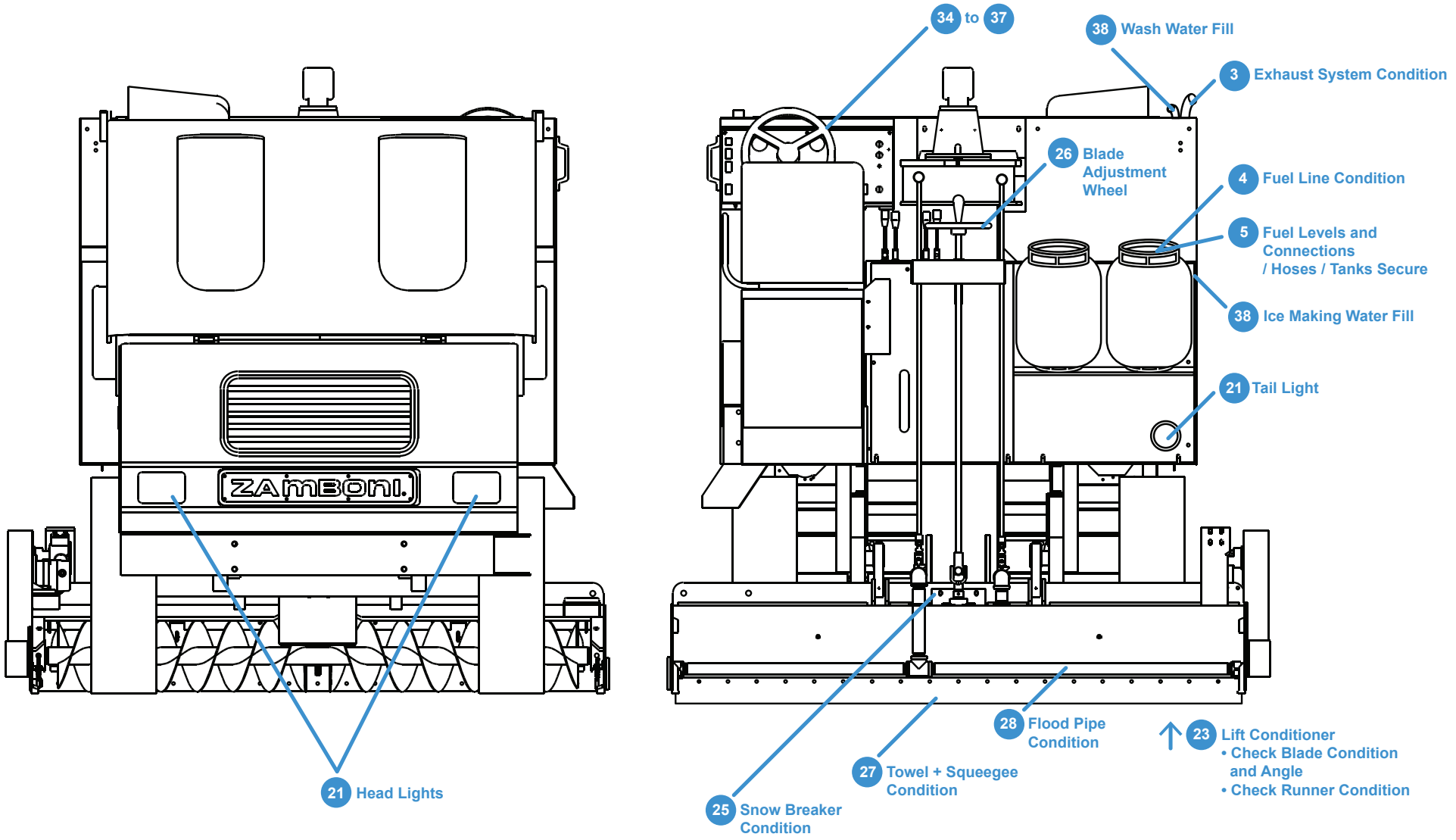
Circle Check Detail:

1. Tank up and snow tank safety stand in place
2. Key out of ignition and secure in your possession
3. Exhaust system condition
4. Fuel line condition
5. Fuel levels and connections/hoses/tanks secure
6. Belts
7. Snow tank seal
8. Hydraulic hoses and couplers
9. Battery and cables
10. Spark plugs
11. Radiator and radiator hoses
12. Air filter
13. Leaf springs
14. Brake lines
15. Fluid levels
16. Hydraulic bypass open or closed
17. U joints
18. Tires (condition and pressure) / hubs / studs / nuts
19. Board brush condition/arm bushing
20. Guide wheel condition
21. Head lights / tail light
22. Safety guards including conditioner guards
23. Lift conditioner to observe blade and runner condition
24. Conditioner / lift arm / bushings
25. Snow breaker condition
26. Blade adjustment wheel
27. Towel and squeegee condition
28. Flood pipe condition
29. Vertical and horizontal auger movement and grease bearings
30. Steering wheel / horn
31. Blade Bar
32. Conditioner leaf springs
33. Auger flighting
34. Hour meter reading
35. Voltmeter
36. Temperature
37. Tachometer
38. Water fill
39. Seat / armrest
40. Poly water tank inspection
41. Snow tank inspection
42. Brake and brake pedal
43. Ice making water and wash water valves
44. Step areas clean and clear
45. Control levers
46. Wash water pump
47. Conveyor drive belt and chain
48. Inspect body condition and safety labels

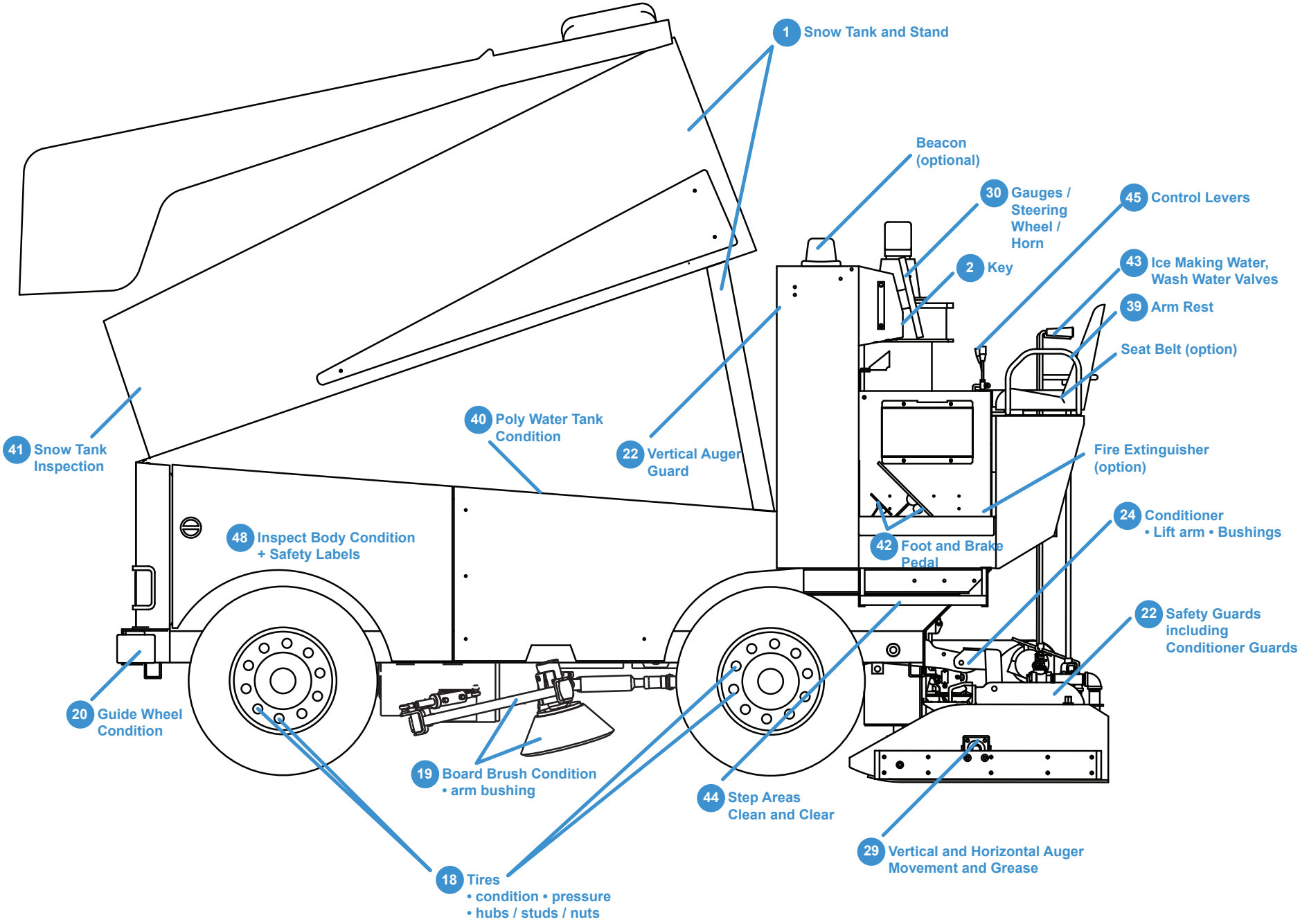
- Exhaust shield (Canadian models)
- Fire extinguisher (option)
- Beacon light (option)
- Backup alarm (option)
- Tire wash water (option)
- Seatbelt (option)

This list is not intended to be a comprehensive checklist and was created as a general guideline to encourage the circle check routine, based on a variety of machine models and features. Your machine model, features and options may vary and those specific items requiring attention should be added to your facility's checklist and circle check routine.

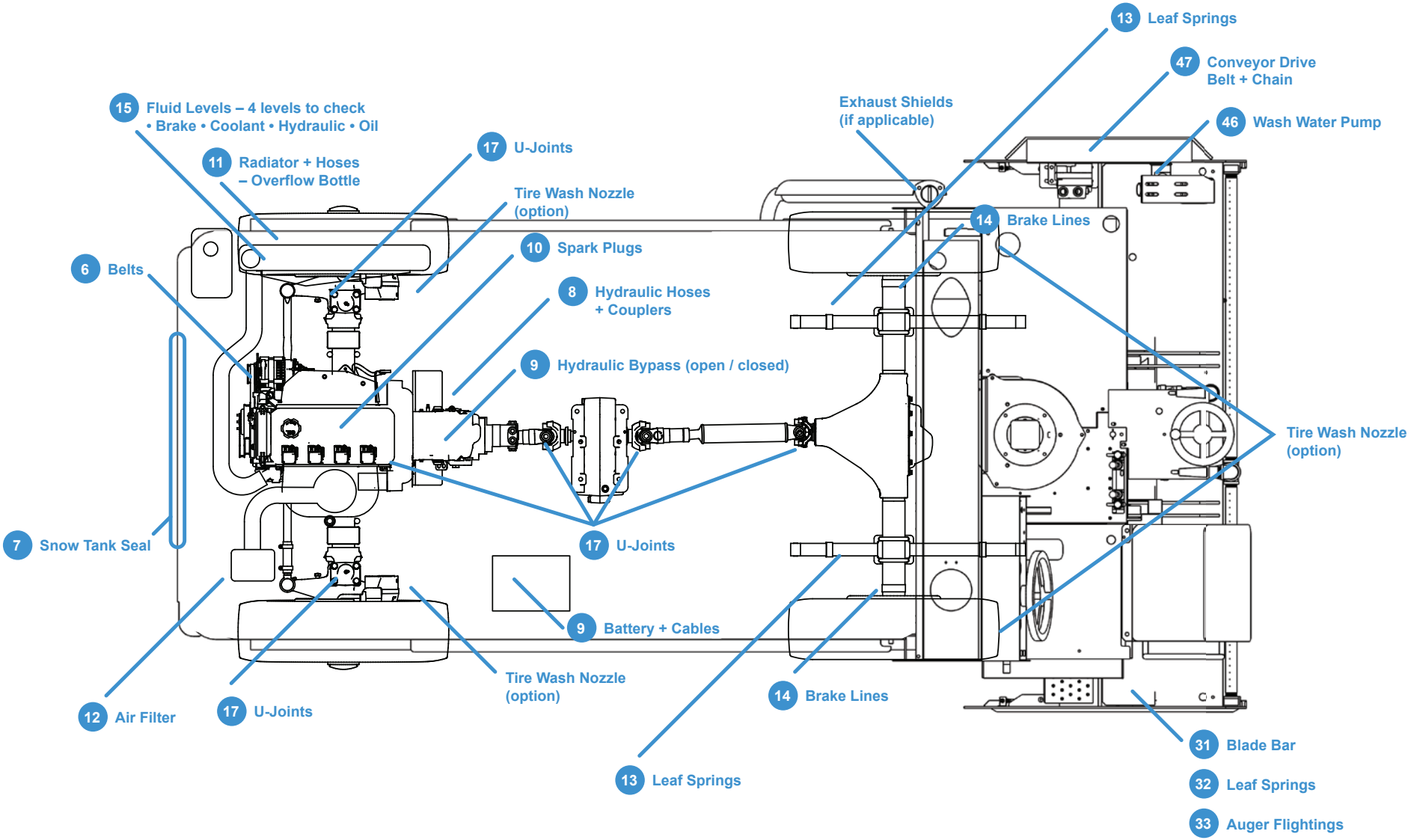
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